A Hybrid AHP–TOPSIS Approach to Laptop Selection
for Office Workers in Vietnam

Group [X] — Logistics Management
University of Finance — Marketing

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Abstract

Selecting a laptop for office work requires balancing several conflicting attributes, and the criteria priorities reported in the international literature do not necessarily transfer to local working conditions. This study develops and applies a decision model for office workers in Vietnam that combines the Analytic Hierarchy Process (AHP) with the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS). In the first stage, candidate criteria were synthesised from prior studies, refined through semi-structured interviews with five experts positioned across the procurement process, and screened through a survey of 84 office workers, retaining seven criteria: price, processor, memory, storage, brand, weight, and battery life. In the second stage, AHP weights were derived from expert pairwise comparisons and verified for consistency ($CR = 0.048$); price (0.349) and processor capability (0.277) jointly account for 62.6% of the total priority, whereas battery life, the most highly rated criterion in the screening survey, receives the smallest weight, reflecting its status as a threshold requirement rather than a differentiator in mains-powered offices. In the third stage, TOPSIS ranked twelve laptop models drawn from seven professional review outlets using market data; the Lenovo ThinkPad T14 Gen 6 (AMD) ranked first ($C_i = 0.6152$), leading on no single attribute but offering the most balanced profile, while the cheapest and the most expensive candidates ranked seventh and last. The model turns an implicit purchase decision into a reproducible and auditable procedure that procurement teams can update as catalogues change.

Keywords: Multi-Criteria Decision-Making; AHP; TOPSIS; laptop selection; office workers; Vietnam.

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Chapter 1

Introduction

1.1 Background and Motivation

The laptop computer has become a basic working tool across most occupations, and its consumption has expanded steadily at the global scale. Statistics compiled by Sönmez Çakır and Pekkaya (2020) indicate that 161.6 million laptops were sold in 2017, roughly 65% more than desktop personal computers. This trajectory was reinforced during the pandemic, when Gaur (2023) reported that worldwide shipments rose to 218 million units in 2020, a 26% increase, and reached 225 million units in 2021. Sustained demand coincides with a relatively short product life cycle, commonly estimated at around five years (Sönmez Çakır & Pekkaya, 2020), which turns device selection into a recurring decision for individuals and organisations alike.

Accompanying this market growth is a rapid fragmentation of the product catalogue. Each generation introduces new configurations and features, increasing the number of brands, models, and variants that a buyer must weigh (Sönmez Çakır & Pekkaya, 2020). Prospective buyers therefore have to compare several technical attributes that are expressed on different scales and that frequently conflict with one another, for instance between processing performance and battery life, or between durability and weight. Under these conditions, relying solely on experience or intuitive comparison offers little guarantee of consistency, because the attributes cannot be reduced directly to a common unit against which trade-offs are made.

A selection problem involving several conflicting attributes falls within the scope of Multi-Criteria Decision-Making (MCDM). The systematic review by Basílio et al. (2022) shows that this family of methods is applied widely across domains, with the Analytic Hierarchy Process (AHP) and the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) among the most frequently used techniques, a pattern also documented through the bibliometric analysis of Zyoud and Fuchs-Hanusch (2017). Combining AHP to derive criteria weights with TOPSIS to rank alternatives is a pairing that recurs in applied research (Tighnavard Balasbaneh et al., 2025), as it separates the estimation of criteria importance from the evaluation of specific alternatives.

1.2 Problem Statement

This study focuses on the situation in which an office worker in Vietnam needs to select a laptop for work. In this setting, price is not fixed in advance but enters as a framing criterion: procurement budgets for office use typically range from 8 to 15 million Vietnamese dong, which bounds the option space before the buyer weighs the remaining hardware specifications. This cost constraint places price in a persistent trade-off with performance, durability, portability, and user experience, so that the final decision depends on how the groups of criteria are balanced rather than on optimising any single indicator.

Although laptop selection has been studied with MCDM tools, the existing body of work reveals two gaps that are relevant to the Vietnamese context. First, most studies on laptop selection rely on AHP or single fuzzy variants to rank criteria (Elnatan & Tannady, n.d.; Sönmez Çakır & Pekkaya, 2020), whereas the integrated AHP–TOPSIS model has been validated mainly on mobile phones (Weng Siew Lam et al., 2023); the application of this pairing to real laptop data from the Vietnamese market remains limited. Second, the criteria priorities obtained in international studies do not transfer fully to local conditions. Battery life, for example, is often identified as the highest-weighted criterion in foreign literature (Weng Siew Lam et al., 2023), yet expert interviews conducted domestically suggest that a range of four to seven hours already meets typical needs given the ready availability of mains power in offices, while device weight receives greater emphasis because of the prevalence of motorbike commuting.

1.3 Research Gap and Contributions

The two limitations outlined above define the space this study occupies. Prior laptop-selection studies have either stopped at criteria weighting without ranking concrete alternatives (Sönmez Çakır & Pekkaya, 2020), or applied the integrated AHP–TOPSIS pairing to product categories other than laptops (Weng Siew Lam et al., 2023); moreover, none has calibrated the criteria set and their priorities to the working conditions of Vietnamese office workers. To address these gaps, this study makes a twofold contribution. First, it constructs a context-specific criteria set for office laptops in Vietnam by combining a literature review with expert interviews, making explicit where local practice diverges from priorities reported in the international literature. Second, it operationalises a complete AHP–TOPSIS decision model on real market data, using AHP to derive criteria weights and TOPSIS to rank actual laptop models, thereby turning the calibrated criteria into an applicable ranking procedure.

1.4 Objectives and Research Questions

The problem outlined above gives rise to three objectives that correspond to three stages of the study. The first stage establishes a set of evaluation criteria suited to Vietnamese office workers, drawing on a literature review combined with expert interviews. The second stage quantifies the relative importance of these criteria using AHP, based on pairwise comparison data elicited from experts. The third stage applies TOPSIS to rank laptop models available on the market according to the derived weights. These objectives are expressed through the following research questions:

1. **RQ1:** Which criteria are appropriate for evaluating laptops for office workers in Vietnam?
2. **RQ2:** What weights do these criteria carry when determined through AHP?
3. **RQ3:** Under a simulated purchase scenario, which laptop model is the most suitable choice under the established criteria and weights when ranked with TOPSIS?

1.5 Scope and Structure of the Report

The study is limited to the office laptop segment relevant to Vietnamese buyers, with product data drawn from Amazon.com listings at a recorded collection date, since a majority of the internationally recommended models considered in Chapter 3 are not yet carried by Vietnamese retail catalogues. The intended beneficiaries of the model are office workers, as distinct from user groups with specialised needs such as graphic design or gaming. The criteria development starts from fourteen candidate criteria grouped into six categories, reflecting economic considerations, core hardware capability, design and interaction experience, portability and durability, ecosystem utility, and trust and after-sales value; a screening survey then narrows this set to the criteria that office workers themselves confirm as important. The study does not seek to generalise across every user segment or distribution channel, but concentrates on the decision situation defined above.

The remainder of the report is organised into five chapters. Chapter 2 reviews the theoretical foundations of MCDM together with the AHP and TOPSIS methods, and surveys prior studies on laptop selection in order to locate the research gap. Chapter 3 presents the research methodology, including the criteria construction process, data collection, and the computational steps. Chapter 4 reports the results on criteria weights and the ranking of alternatives. Chapter 5 discusses the implications of these results in relation to the existing literature, and Chapter 6 summarises the contributions and limitations of the study.

Chapter 2

Literature Review

2.1 Laptop Selection as a Multi-Criteria Decision Problem

Choosing a laptop requires weighing attributes that cannot be reduced to a single scale. Processing performance, battery life, durability, portability, and price move in different directions and often trade off against one another: a lighter chassis tends to shrink battery capacity, and a lower price tends to constrain processing power. Sönmez Çakır and Pekkaya (2020) frame this directly, noting that new features are added to laptops every year and that the resulting variety of brands, models, and configurations forces buyers into a comparison problem that intuition alone cannot resolve consistently. Gaur (2023) reach a similar conclusion from the demand side: as shipments grew sharply during the pandemic, the number of criteria buyers had to reconcile, from brand and price to processor and screen size, grew with it. A decision problem of this kind, where several conflicting criteria must be evaluated jointly against a set of alternatives, falls within the scope of Multi-Criteria Decision-Making (MCDM), the methodological family this chapter now turns to.

2.2 An Overview of MCDM Methods

MCDM research has grown into an established branch of operations research. Based on a Scopus bibliometric analysis, Zyoud and Fuchs-Hanusch (2017) report 10,188 documents on AHP and 2,412 documents on TOPSIS through 2015, with China producing the largest national share of each (34.5% and 35.1%, respectively) and *Expert Systems with Applications* standing out as the most productive outlet for both techniques. This scale of activity reflects a broad family of methods rather than a single dominant technique.

Villalba et al. (2024) organise this family into five groups according to the mechanism each uses to arrive at a preference. Direct scoring methods, such as the Simple Additive Weighting technique and its extension COPRAS, aggregate weighted performance scores directly. Distance-based methods, principally TOPSIS and VIKOR, rank alternatives by their geometric distance to an ideal solution. Pairwise comparison methods, led by AHP together with the Analytic Network Process and MACBETH, derive priorities from relative judgements between pairs of criteria or alternatives. Outranking methods, such as PROMETHEE and ELECTRE, build a preference relation from the degree to which one alternative outranks another rather than from a single aggregated score. Utility and value methods, including MAUT, MAVT, and MIVES, construct explicit utility functions over each criterion. This taxonomy is useful because it shows that AHP and TOPSIS belong to different groups, pairwise comparison and distance-based respectively, so that combining them addresses two distinct parts of the decision problem, weighting and ranking, rather than duplicating the same mechanism twice.

The prevalence of these two techniques is confirmed by counts of published applications. Basílio et al. (2022), reviewing 23,494 documents published between 1977 and 2022, record 6,835 applications of AHP and 4,907 of TOPSIS, ahead of VIKOR (1,475), PROMETHEE (1,382), ANP (1,262), ELECTRE (1,005), and DEMATEL (888). Within their own domain, Villalba et al. (2024) report a comparable pattern within their own domain: the distance-based and pairwise comparison groups identified above are the two most frequently applied, AHP alone accounts for 35% of the reviewed articles and its combination with TOPSIS for a further 31%, ahead of the next most common technique, Simple Additive Weighting, at 11%, and AHP overall, whether used alone or hybridised with another technique, appears in 71% of the sample. Hybridisation with another MCDM method occurs in 55 of the reviewed articles, while only 5 propose a modification to an existing method rather than applying it as originally formulated, indicating that researchers in this domain favour combining established techniques over developing new ones. Tighnavard Balasbaneh et al. (2025) likewise observe that combining AHP and TOPSIS is used more often than other method pairings, although they caution that no single combination has emerged as a settled consensus across application domains. Taken together, these figures indicate that AHP and TOPSIS are not only individually common but are also the pairing most frequently chosen when a study needs both criteria weights and an alternative ranking, which motivates a closer look at each method in turn.

2.3 The Analytic Hierarchy Process

The Analytic Hierarchy Process was introduced by Saaty (1987) as a method of measurement built on ratio scales derived from pairwise comparison. According to Zyoud and Fuchs-Hanusch (2017), AHP rests on three principles: decomposition, which structures a decision into a hierarchy of a goal, criteria, and alternatives; comparative judgement, which elicits the relative importance of elements at the same hierarchical level through pairwise comparison; and synthesis of priorities, which aggregates these judgements into an overall priority vector. Comparisons are expressed on a nine-point ratio scale (Table 3.1), which allows both quantitative and qualitative criteria to be evaluated within the same framework, a property that makes AHP well suited to decisions such as laptop selection in which some attributes, for instance processor performance, are naturally numeric while others, such as brand perception, are not.

Because the judgements underlying a pairwise comparison matrix are elicited from a human respondent rather than measured directly, AHP incorporates an explicit check on their internal coherence. The consistency ratio, obtained from the principal eigenvalue of the comparison matrix as detailed in Section 3.3 (Equation (3.3)) and formalised by Saaty (2008), flags matrices in which the respondent's judgements contradict one another, for example by rating criterion A more important than B, B more important than C, yet C more important than A. This consistency check is one of the reasons AHP remains attractive for eliciting subjective expert weights: unlike a simple ranking or scoring exercise, it provides a quantitative signal for when a judgement set should be revisited. The method is not without limitations, however. Zyoud and Fuchs-Hanusch (2017) note that AHP is susceptible to rank reversal, whereby the relative ranking of two existing alternatives can change when a new, unrelated alternative is added to the comparison, and that the pairwise comparison procedure assumes criteria are independent of one another, an assumption that does not always hold in practice.

Where expert judgements are too imprecise to be expressed as single crisp numbers, several studies extend AHP into a fuzzy environment. Liu et al. (2020) organise this literature around four technical choices that a fuzzy AHP model must make: how the relative importance of a comparison is represented, typically as a fuzzy number rather than a crisp value; how judgements from multiple experts or multiple criteria are aggregated; how the resulting fuzzy priorities are defuzzified back into crisp weights for practical use; and how consistency is measured once judgements are no longer single numbers. This layer of fuzzification is warranted when experts cannot commit to a precise value, for instance under significant uncertainty about future conditions. In the present study, criteria weights are elicited from an expert with direct, first-hand purchasing experience, whose pairwise judgements can be expressed as ordinary Saaty-scale values without this additional imprecision; the method therefore proceeds with the crisp form of AHP described in Section 3.3, while the consistency safeguard that AHP provides remains in place.

2.4 The TOPSIS Method

The Technique for Order of Preference by Similarity to Ideal Solution was proposed by Hwang and Yoon (1981) on the principle that the best alternative should lie simultaneously closest to a positive ideal solution and farthest from a negative ideal solution, where the positive ideal solution combines the best attainable value on every criterion and the negative ideal solution combines the worst. Behzadian et al. (2012) note that this requires attribute values that are numeric, monotonically increasing or decreasing in desirability, and measured on commensurable scales, conditions that a laptop dataset of prices, benchmark scores, and physical measurements satisfies directly.

Reviewing 266 applications of TOPSIS published across 103 journals since 2000, Behzadian et al. (2012) find the method concentrated in supply chain management and logistics (27.5% of the sample) and in design, engineering, and manufacturing systems (23%), together accounting for more than half of the applications surveyed. The same review attributes this spread to several practical advantages: TOPSIS makes full use of the available attribute information rather than discarding it at an intermediate step, produces a cardinal ranking of every alternative rather than a partial order, does not require the criteria to be preferentially independent, and remains conceptually and computationally simple regardless of how many criteria or alternatives are involved. Zyoud and Fuchs-Hanusch (2017) similarly place TOPSIS as the second most frequently used MCDM technique after AHP, consistent with the frequency counts reported in Section 2.2.

As with AHP, a fuzzy extension exists for settings in which attribute values themselves are imprecise rather than directly measurable. Chen (2000) extend TOPSIS to a fuzzy environment by representing ratings and weights as triangular fuzzy numbers, computing distances to a fuzzy positive and negative ideal solution through the vertex method, and deriving a closeness coefficient from these distances. This extension is most useful when performance data are themselves elicited as linguistic judgements. The laptop dataset used in this study, in contrast, is built from directly observable specifications and market prices, so the crisp TOPSIS procedure described in Section 3.4 is applied without a fuzzy layer.

2.5 Prior Studies on Laptop and Electronics Selection

Applying MCDM to laptop and related electronics choices is an established but fragmented line of research, summarised in Table 2.1.

Three studies apply MCDM directly to laptop selection. Sönmez Çakır and Pekkaya (2020) combine DEMATEL, used to map how criteria influence one another, with fuzzy AHP, used to derive their priorities, and find that brand image carries the highest weight among six main criteria, together with price and peripheral properties accounting for roughly two-thirds of the total priority. Gaur (2023) weight nine criteria with a joint AHP and Best-Worst Method procedure and rank nine laptop brands with TOPSIS, identifying Dell as the top-ranked alternative during the pandemic period they study. Working with a smaller, price-sensitive segment, Elnatan and Tannady (n.d.) apply AHP to laptop choice among university students in North Jakarta and find that price, processor, and RAM together account for the largest share of the priority weight, ahead of graphics capability, storage, and brand.

Two further studies apply MCDM to closely related device-selection problems. Guan and Yuen (2022) propose a cognitive comparison rating combined with hierarchical clustering for personalised laptop recommendation, and benchmark it explicitly against AHP-style pairwise rating. Pramanik et al. (2021) compare several MCDM techniques, including TOPSIS, for selecting computing resources in a mobile crowd-computing setting, with attention to how stable the resulting rankings remain across methods.

A separate body of consumer-behaviour research does not apply MCDM but supplies evi-

Table 2.1: Prior studies on laptop and electronics selection using MCDM.

Source	**Product / Area**	**Criteria (illustrative)**	**Method**
Sönmez Çakır & Pekkaya (2020)	Laptop	Price, speed, storage, monitor, peripherals, brand	DEMATEL + Fuzzy AHP
Gaur (2023)	Laptop (COVID-19)	Brand, price, storage, RAM, processor, weight, screen	AHP + BWM + TOPSIS
Elnatan & Tannady	Laptop (students)	Brand, processor, VGA, RAM, storage, price	AHP
Lam et al. (2023)	Mobile phone	Technical spec, physical properties, user-related features	AHP + TOPSIS
Maghsoudi et al. (2026)	Notebook	Price, design, display, CPU, RAM, battery	Sentiment mining + Fuzzy + TOPSIS
Rau & Fang (2018)	Notebook (purchase timing)	Functional value, brand value, cost, eco-efficiency	TOPSIS
Liao & Chuang (2022)	Eco-friendly laptop	Price, battery, shell material, CPU, monitor	Conjoint analysis
Kang et al. (2022)	Laptop (online reviews)	Appearance, configuration, logistics, after-sales	Empirical review mining
Guan & Yuen (2022)	Laptop recommendation	Product attributes vs. user preference	Pairwise rating + clustering
Pramanik et al. (2021)	Mobile device selection	Computing resource parameters	Multi-method MCDM comparison
Šostar & Ristanović (2023)	Consumer behaviour	Cultural, social, personal, psychological factors	AHP
Jiménez-Parra et al. (2014)	Remanufactured laptop	Attitude, subjective norm, motivation	Consumer-behaviour model

Source: authors' own compilation from the studies reviewed in this section.

dence on which criteria matter and how they are weighted in practice, complementing the criteria identified through the expert interviews in Section 3.2.2. Using conjoint analysis, Liao and Chuang (2022) find that price is the single most influential attribute in eco-friendly laptop choice, ahead of battery life, shell material, and processor. Mining online reviews on a large Chinese retail platform, Kang et al. (2022) show that purchase decisions are also shaped by service-related factors, appearance design, logistics service, and after-sales support, that fall outside the hardware specifications most MCDM studies emphasise. Rau and Fang (2018) apply TOPSIS to identify the optimal time to purchase a notebook once functional value, brand value, cost, and eco-efficiency are weighed jointly, while Šostar and Ristanović (2023) use AHP on a broader consumer-behaviour sample and find personal budget the most influential factor among cultural, social, personal, and psychological drivers of purchase, a result consistent with the central role assigned to price throughout this literature.

These studies collectively establish that laptop and electronics selection is a recurring MCDM application and that price, processing capability, and, increasingly, service-related criteria dominate the resulting priority structures. None of them, however, combines a full AHP–TOPSIS pipeline with laptop-specific market data collected in Vietnam, and none tests whether the priority structures they report, largely derived from markets outside Southeast Asia, hold under local working conditions.

2.6 Research Gap and Positioning

Two gaps follow from this review, giving fuller evidential grounding to the limitations anticipated in Section 1.3. First, among the studies that address laptop selection directly, those relying on AHP alone or on fuzzy variants stop at criteria weighting without ranking concrete alternatives (Sönmez Çakır & Pekkaya, 2020), while the integrated AHP–TOPSIS pipeline has been validated mainly on other product categories, such as mobile phones (Weng Siew Lam et al., 2023); its application to real laptop market data remains comparatively rare. Second, the priority structures reported in this literature, for instance the dominance of brand image (Sönmez Çakır & Pekkaya, 2020) or of price (Liao & Chuang, 2022), are derived from markets and user populations outside Vietnam, leaving open whether they transfer to the working conditions of Vietnamese office employees. This positions the present study within the pairing that Section 2.2 identifies as the most frequently applied, AHP for criteria weighting and TOPSIS for ranking (Tighnavard Balasbaneh et al., 2025), while directing it at the specific combination of product, market, and user population that the reviewed literature has not yet covered.

Chapter 3

Methodology

3.1 The Proposed Research Model

The study is organised into three sequential stages, each of which answers one of the research questions stated in Section 1.4. The first stage develops the set of evaluation criteria: candidate criteria are synthesised from the literature reviewed in Chapter 2, refined through expert interviews, and then screened through a large-scale survey, so that only criteria whose importance is confirmed by actual office workers enter the quantitative model. The second stage quantifies the relative importance of the retained criteria with AHP, including an explicit consistency check before the weights are accepted. The third stage applies the model to a concrete purchase scenario: candidate laptops are identified from the recommendation lists of professional review outlets, their attribute data are collected from the Vietnamese retailer CellphoneS and operationalised into a decision matrix, and TOPSIS ranks them using the weights obtained in the second stage. The stages are strictly ordered: the pairwise comparison in Stage 2 cannot be designed until the criteria set is fixed, and the ranking in Stage 3 is meaningful only after the weights have passed the consistency check. Figure 3.1 summarises this pipeline.

3.2 Stage 1 — Criteria Development

3.2.1 Literature Synthesis

The starting point of the criteria set is the body of laptop-selection and consumer-behaviour studies reviewed in Section 2.5. From these sources, every attribute that at least one prior study used to evaluate laptops or closely related devices was extracted and consolidated: price appears in nearly all of them (Liao & Chuang, 2022; Rau & Fang, 2018; Sönmez Çakır & Pekkaya, 2020; Weng Siew Lam et al., 2023), core hardware attributes such as processor, memory, and storage recur across the MCDM applications (Gaur, 2023; Maghsoudi et al., 2026), and service-related attributes such as brand and after-sales support are emphasised in the review-mining and consumer-behaviour literature (Kang et al., 2022; Šostar & Ristanović, 2023). Attributes

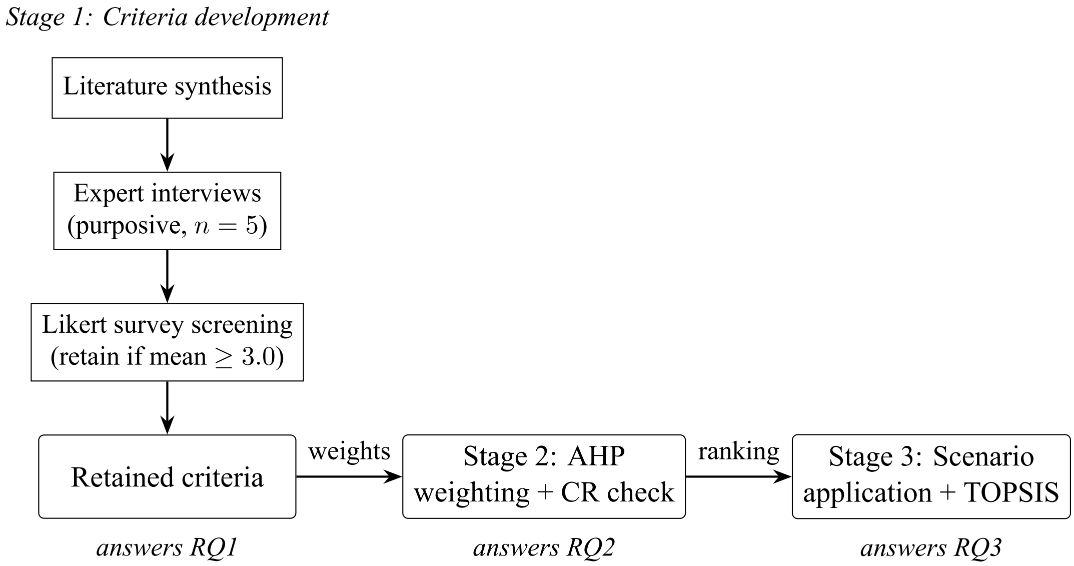

Figure 3.1: The proposed research model.

Source: authors' own elaboration.

that overlap in meaning were merged into a single criterion, and attributes tied to specialised use cases outside the office-worker persona, such as colour-accurate displays for design work, were excluded at this stage. The remaining attributes were then organised into thematic groups reflecting different aspects of the purchase decision, such as core hardware capability or trust and after-sales value, so that the pairwise comparisons of Section 3.3 could later be structured hierarchically rather than as a single flat list. The resulting set of candidate criteria is reported in Section 4.1.1.

3.2.2 Expert Interviews

Because the candidate criteria from Section 3.2.1 were derived from studies conducted outside Vietnam, they were validated against local practice before being quantified. Five experts were selected purposively so that each occupies a different position in the laptop procurement process: a deputy head of an international relations office with seven years of end-user experience (CG-01), an IT administrator who specifies and maintains office fleets (CG-02), an import–export documentation officer representing the target persona (CG-03), a head of product development with ten years of managerial purchasing experience (CG-04), and a retail sales consultant who advises laptop buyers daily (CG-05).

The interview protocol itself, rather than being improvised, was constructed following the Interview Protocol Refinement (IPR) framework, whose four phases require that every question be traceable to a research objective, that the questions be sequenced as an inquiry-based conversation rather than a checklist, that the draft be reviewed before fieldwork, and that it be piloted (Castillo-Montoya, 2016). Applying this framework produced a five-part funnel that moves from the concrete to the general: a warm-up section grounding each respondent in a recent, specific laptop-selection episode; a stimulus-based section in which a literature finding for each criterion group is read aloud immediately before the matching question, so that agreement or disagreement is anchored to a stated claim rather than solicited as an abstract opinion; an open section probing for criteria specific to the Vietnamese context that the literature synthesis could not have anticipated; a trade-off scenario that generates qualitative seed data for the pairwise comparisons of Section 3.3; and a closing section built around member-checking, in which the interviewer restates the respondent's positions for confirmation. Most substantive questions in this structure carry a follow-up probe that asks the respondent to recall the specific case behind an answer or to explain the reasoning process that produced it, a technique adapted from cognitive interviewing that keeps a response anchored in the respondent's actual experience rather than a general impression formed on the spot (Willis, 2005). This funnel structure and probing technique operate inside a semi-structured session that functions as the qualitative first round of a Delphi-style refinement preceding AHP, an arrangement applied in prior criteria-development work (Al Hazza et al., 2022; McMillan et al., 2016). Each session lasted twenty to thirty minutes; all five respondents gave informed consent, no audio was recorded, and only written notes were retained.

Interview notes were reviewed for proposals to add, remove, or redefine a criterion. To keep the revision process auditable rather than ad hoc, a proposal was adopted only when it was corroborated independently by a majority of the five experts, and any resulting change to the candidate set was cross-checked against the literature reviewed in Section 2.5 before being finalised. The outcome of this revision process is reported together with the resulting criteria set in Section 4.1.1.

3.2.3 Survey-Based Screening

Expert validation establishes that a criterion is defensible, but not that the target population actually weighs it when buying. The candidate criteria from Section 3.2.2 were therefore submitted to a wider screening survey before any pairwise comparison was designed. A questionnaire was distributed through Google Forms in which respondents rated the importance of each criterion when choosing a laptop for office work on a five-point Likert scale, from 1 (not important) to 5 (very important). The instrument deliberately contained one rating item per candidate criterion and no other items, keeping completion time short and avoiding demographic questions that the screening decision does not use.

The retention rule was fixed before data collection: a criterion is retained if its sample mean is at least 3.0, the midpoint of the scale, and removed otherwise. On a five-point scale the midpoint separates responses that lean towards importance from those that lean away from it, so a mean below 3.0 indicates that respondents on balance do not regard the criterion as important (Chyung et al., 2017). Committing to the threshold in advance prevents the rule from being adjusted after the results are known. Responses missing more than 20% of the items were to be discarded, and straight-lined responses, in which a respondent gives the same rating to every item, were to be flagged for a group decision rather than silently removed. The screening outcome, which reduces the candidate set to the criteria carried into Stage 2, is reported in Section 4.2.

3.3 Stage 2 — AHP for Criteria Weighting

The relative importance of the criteria retained after screening is derived through the Analytic Hierarchy Process, whose theoretical basis is reviewed in Section 2.3. The procedure follows five steps.

Step 1 — Hierarchy construction. The decision problem is structured into three levels: the goal of selecting the most suitable laptop for a Vietnamese office worker at the top, the criteria retained from Stage 1 at the intermediate level, and the candidate laptop models at the bottom. Only the upper two levels are involved in this stage; the alternatives enter the model in Stage 3, where they are evaluated on directly measurable data rather than on pairwise judgements.

Step 2 — Pairwise comparison. The pairwise judgements were elicited from one representative expert drawn from the interview panel of Section 3.2.2, selected for direct, first-hand experience of office laptop procurement. For n retained criteria the expert completes $n(n-1)/2$ comparisons, each expressing how much more important one criterion is than another on the nine-point ratio scale of Table 3.1. Relying on a single judgement set concentrates the weighting in one perspective, which is why the consistency check in Step 5 is applied strictly: an inconsistent matrix is returned to the expert for re-evaluation rather than accepted or averaged away.

Table 3.1: The Saaty fundamental scale of pairwise comparison.

Intensity	Definition
1	Equal importance
3	Moderate importance
5	Strong importance
7	Very strong importance
9	Extreme importance
2, 4, 6, 8	Intermediate values

Source: adapted from Saaty (1987).

Step 3 — Comparison matrix. For n criteria, the pairwise judgements form a reciprocal matrix $A = [a_{ij}]$ with $a_{ij} = 1/a_{ji}$ and $a_{ii} = 1$:

$$A = \begin{bmatrix} 1 & a_{12} & \cdots & a_{1n} \\ 1/a_{12} & 1 & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 1/a_{1n} & 1/a_{2n} & \cdots & 1 \end{bmatrix}. \tag{3.1}$$

Only the $n(n-1)/2$ entries above the diagonal are elicited; the remaining entries follow from reciprocity, so the expert is never asked the same comparison twice.

Step 4 — Priority (weight) vector. The criteria weights are taken as the principal right eigenvector of A, that is, the vector **w** satisfying

$$A\mathbf{w} = \lambda_{\text{max}}\mathbf{w}, \qquad \sum_{i=1}^{n} w_i = 1, \tag{3.2}$$

where λ_{max} is the largest eigenvalue of A. Several simpler approximations circulate in applied work, including the normalised column average used by Weng Siew Lam et al. (2023); the eigenvector is preferred here because Saaty (2003) shows it to be the theoretically required solution whenever the judgement matrix is a small perturbation of a fully consistent one, which the consistency check in Step 5 verifies, and because the comparative simulations of Ishizaka and Lusti (2006) find that approximation methods diverge from the eigenvector increasingly as matrices grow larger or less consistent. In computation the eigenvector is obtained by power iteration, repeatedly multiplying a normalised weight vector by A until convergence. As a cross-check, the weights are also computed by the geometric mean of each row; agreement between the two vectors in the ordering of criteria is treated as evidence that the result does not depend on the derivation method.

Step 5 — Consistency check. Because the judgements are elicited from a human respondent, they may contradict one another, and the eigenvector of a heavily contradictory matrix is not a meaningful weight vector. The internal coherence of the judgements is therefore verified through the consistency ratio $CR = CI/RI$, where the consistency index is

$$CI = \frac{\lambda_{\text{max}} - n}{n-1}, \tag{3.3}$$

λ_{max} is the principal eigenvalue from Equation (3.2), and RI is the random index, the mean consistency index of randomly generated reciprocal matrices of the same order (Table 3.2). A matrix is accepted when $CR \leq 0.10$ (Saaty, 2008). If the threshold is exceeded, the weights are not used: the three pairwise entries that deviate most from the values a consistent matrix would imply are identified and returned to the expert for re-evaluation, and the matrix is recomputed. The threshold is never relaxed to accommodate a marginally failing matrix.

Table 3.2: Random Index (RI) values by matrix order n.

n	3	4	5	6	7	8	9	10
RI	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49

Source: adapted from Saaty (1987).

3.4 Stage 3 — Scenario-Based Application with TOPSIS

3.4.1 Decision Scenario

The third stage applies the weighted criteria to a concrete decision situation. The scenario mirrors the problem statement of Section 1.2: an office worker in Vietnam needs a laptop for work, has browsed review websites and the catalogues of established retailers, and still cannot decide, because the information available compares products one attribute at a time while the actual decision requires weighing all attributes at once. Framing Stage 3 as a scenario serves two purposes. It keeps the demonstration honest, since every modelling choice, from which laptops enter the comparison to how each criterion is measured, must be one this buyer could reproduce from publicly available information; and it makes the output directly interpretable, since the resulting ranking answers the question the buyer actually asks. The stage proceeds in three steps that parallel a standard data workflow: identifying the alternatives, collecting and operationalising their attribute data, and analysing the resulting decision matrix with TOPSIS.

3.4.2 Alternative Identification

The candidate laptops are not chosen by the researchers' own judgement, which would risk preselecting the winner, but harvested from published recommendation lists of professional review outlets and then filtered for local availability. An outlet is included in the source pool if it satisfies three conditions: it maintains a documented, standardised laboratory testing protocol rather than opinion-only reviews; it actively covers laptops as a product category; and its editorial operation is stable enough that its current recommendations reflect that protocol. Seven outlets meet these conditions, summarised in Table 3.3. One widely known outlet, Engadget, was considered and excluded: it reduced its editorial staff by roughly one third in 2024, reoriented its output towards search-driven affiliate content, and changed ownership in early 2026, so the stability condition is not met. CNET is retained with a caveat: the outlet published AI-generated articles without clear disclosure in 2023, but it operates its own testing laboratory and tightened human editorial review after the incident, and its current laptop recommendations are lab-based.

Table 3.3: Review outlets used as sources for alternative identification.

Outlet	Operator	Testing approach
PCMag	Ziff Davis	Standardised lab benchmarks since 1982; Editors' Choice programme
Tom's Guide	Future plc	Comparative product testing; seasonal category rankings
TechRadar	Future plc	Global review team; per-segment rankings (business, budget)
Laptop Mag	Future plc	Laptop-only coverage; in-house battery and performance benchmarks
CNET	Ziff Davis	In-house lab for performance, battery, and display testing
Wirecutter	The New York Times	Hands-on testing; no sponsored review placements
RTINGS	Independent	Instrumented measurement (display, thermals, input latency)

Source: authors' own compilation from the editorial policies of the listed outlets.

From each outlet in the pool, the current recommendation list for the best business or premium laptops of the model year is collected, and the union of these lists forms the raw candidate set. Because the same model frequently appears on several lists, the set is deduplicated by manufacturer and model designation, treating configuration variants of the same model as a single candidate at this step. Price and configuration data for the deduplicated set are then collected from Amazon.com rather than from a Vietnamese retailer, because the outlets' recommendations are dominated by newly launched, internationally released configurations that Vietnamese retail catalogues do not yet carry at the time of collection, so the buyer of the scenario in Section 3.4.1 would in practice have to source most of these models through an international retailer regardless of the Vietnamese market context. Amazon.com is the world's largest online retailer and the leading e-commerce platform in the United States, and using its product listings as a source of laptop specification and price data has direct precedent in the consumer-electronics literature: Rau and Fang (2018) construct their notebook price-history dataset directly from Amazon product listings. The date of collection is recorded with the data, since both the recommendation lists and the retailer's catalogue change over time.

3.4.3 Data Collection and Operationalisation

Each remaining alternative is scored on the retained criteria, and the way each criterion is measured is fixed before the data are collected. Five criteria are directly quantitative and are taken as listed on Amazon.com: the price in US dollars, the memory capacity in gigabytes, the storage capacity in gigabytes, the device weight in kilograms (converted from the pounds figure shown in the listing), and the battery capacity in watt-hours as specified by the manufacturer. Watt-hours are used in preference to a manufacturer-claimed runtime in hours, because the latter is measured under differing, vendor-specific workloads and is therefore not commensurable across brands, whereas watt-hour capacity is a fixed physical specification of the battery itself. The processor cannot be treated this way: chipset designations such as product names and generation labels are neither numeric nor directly comparable across manufacturers. The criterion is therefore operationalised through a standardised benchmark score, using the PassMark CPU database, which converts heterogeneous processor names into a single performance scale. This step matters for the validity of the analysis, because TOPSIS requires attribute values that are numeric, monotonic in desirability, and commensurable (Behzadian et al., 2012); a benchmark score satisfies all three conditions where a chipset name satisfies none. Brand, the one qualitative criterion, is likewise converted into an observable quantity rather than scored by hand: it is operationalised as each manufacturer's worldwide PC shipment market share for the first quarter of 2026, as reported by the market-research firm Gartner (Gartner, Inc., 2026). This proxy is chosen in preference to a Vietnam-specific measure, such as a count of authorised service centres, because no freely accessible source publishes brand-level market share for the Vietnamese laptop market specifically; global shipment share is disclosed quarterly and is independently reproducible, and it still functions as the concrete, verifiable expression of manufacturer standing that the trust and after-sales group of Table 4.1 calls for, whereas a hand-assigned brand score would reintroduce exactly the subjectivity that the screening and weighting stages were designed to control. Gartner is used in preference to IDC, the other major PC-market tracker, because Gartner's quarterly release reports a share figure for every one of the six brands present in this candidate set, including Acer, whereas IDC's release names only its top five vendors by shipment volume and groups every remaining brand, Acer included, into a single undifferentiated "other vendors" residual; using a single source that individually reports all six brands avoids mixing two trackers' figures within the same criterion column.

3.4.4 The TOPSIS Procedure

Given the criteria weights from Stage 2 and the decision matrix assembled above, the candidate laptops are ranked with TOPSIS over seven steps.

Step 1 — Decision matrix. For m alternatives and n criteria, $X = [x_{ij}]$ collects the performance of alternative i on criterion j.

Step 2 — Vector normalisation.

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}}. \tag{3.4}$$

Step 3 — Weighted normalised matrix.

$$v_{ij} = w_j \, r_{ij}. \tag{3.5}$$

Step 4 — Ideal solutions. The positive ideal solution A^+ and negative ideal solution A^- are formed from the best and worst weighted scores of each criterion, respecting its benefit or cost direction:

$$A^+ = \{v_1^+, \dots, v_n^+\}, \qquad A^- = \{v_1^-, \dots, v_n^-\}. \tag{3.6}$$

Step 5 — Separation measures.

$$S_i^+ = \sqrt{\sum_{j=1}^{n} \left(v_{ij} - v_j^+\right)^2}, \qquad S_i^- = \sqrt{\sum_{j=1}^{n} \left(v_{ij} - v_j^-\right)^2}. \tag{3.7}$$

Step 6 — Closeness coefficient.

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}, \qquad 0 \le C_i \le 1. \tag{3.8}$$

Step 7 — Ranking. The alternatives are ranked in descending order of C_i; the laptop with the highest C_i is the recommended choice.

Chapter 4

Results

4.1 Criteria Development Outcomes

4.1.1 The Candidate Criteria Set

Applying the literature-synthesis procedure of Section 3.2.1 to the studies reviewed in Section 2.5 produced fourteen candidate criteria. These were organised into six groups reflecting the aspects of the purchase decision they describe: economic cost, core hardware capability, design and interaction experience, portability and durability, ecosystem utility, and trust and after-sales value. Table 4.1 lists the resulting set together with each criterion's group and its direction of preference. Most criteria are benefit criteria, for which a larger value is preferred; price and physical size or weight are cost criteria, for which a smaller value is preferred. The distinction matters at two later points: it determines the direction of the ideal solutions in TOPSIS (Section 3.4), and it guards against the error of treating a heavier or more expensive machine as better simply because its raw attribute value is larger.

4.1.2 Interview-Driven Refinement

Following the revision rule described in Section 3.2.2, the interviews changed the candidate set in two ways. First, the dedicated graphics card was removed. Four of the five experts converged independently on the argument that a discrete GPU is a cost without a corresponding benefit for office work: it raises the purchase price, weight, and thermal load while shortening battery life, and the budget it consumes is better redirected towards a newer-generation processor, more memory, or faster storage. This judgement is consistent with the finding of Maghsoudi et al. (2026) that the graphics card ranked last among ten notebook attributes in their expert-weighted model. Second, software compatibility was added. The experts who work inside Vietnamese offices (CG-01, CG-03) described a lock-in effect that the international literature rarely makes explicit: accounting packages, internal ERP systems, and the government's customs declaration software are written for Windows, so a device that cannot run them is excluded from considera-

Table 4.1: The fourteen candidate evaluation criteria in six groups.

ID	Criterion	Type	Group
C1	Price	Cost	Economic
C2	CPU	Benefit	Core hardware
C3	RAM	Benefit	Core hardware
C4	Storage (SSD)	Benefit	Core hardware
C5	Display	Benefit	Design & interaction
C6	Keyboard	Benefit	Design & interaction
C7	Design & shell material	Benefit	Design & interaction
C8	Size / weight	Cost	Portability & durability
C9	Durability	Benefit	Portability & durability
C10	Battery life	Benefit	Portability & durability
C11	Connectivity ports	Benefit	Ecosystem utility
C12	Software compatibility	Benefit	Ecosystem utility
C13	Brand	Benefit	Trust & after-sales
C14	After-sales / warranty	Benefit	Trust & after-sales

Source: authors' own synthesis from the reviewed literature and the expert interviews.

tion regardless of its other merits. The interviews also sharpened the definitions of two retained criteria, restricting the storage criterion to solid-state drives and extending the design criterion to cover shell material, and recalibrated the expectation for battery life: because mains power is readily available in offices, a runtime of four to seven hours was considered sufficient, a markedly lower bar than the top-priority status that battery life receives in Weng Siew Lam et al. (2023). The revised set therefore still contains fourteen criteria, but with GPU replaced by software compatibility and with definitions anchored to local working conditions; this is the candidate set reported in Table 4.1 above and carried into the screening survey of Section 3.2.3.

4.2 Criteria Screening Results

The screening survey described in Section 3.2.3 collected 84 responses, all of which were complete: no respondent left more than 20% of the items unanswered, and no straight-lined response was detected, so every questionnaire entered the analysis. Table 4.2 reports the descriptive statistics for the fourteen candidate criteria together with the retention decision under the pre-committed threshold of a mean of at least 3.0.

Seven criteria pass the threshold: battery life records the highest mean (4.02), followed by memory and physical size or weight (both 3.96), storage (3.89), processor (3.86), and price and brand (both 3.79). The seven removed criteria all fall clearly below the midpoint, from keyboard experience (2.88) down to durability (2.36). The gap between the lowest retained mean (3.79) and the highest removed mean (2.88) spans almost a full scale point, so the composition of the retained set is not sensitive to small movements of the threshold around the midpoint. These seven criteria constitute the answer to RQ1 and define the dimension of the pairwise comparison

Table 4.2: Descriptive statistics and screening decision for the candidate criteria.

Criterion	N	Mean	SD	Decision
Price	84	3.79	1.27	Retain
CPU	84	3.86	1.13	Retain
RAM	84	3.96	1.02	Retain
SSD	84	3.89	1.12	Retain
Display	84	2.80	1.32	Remove
Keyboard	84	2.88	1.07	Remove
Design & shell material	84	2.83	1.16	Remove
Size / weight	84	3.96	1.02	Retain
Durability	84	2.36	1.42	Remove
Battery life	84	4.02	0.97	Retain
Connectivity ports	84	2.67	1.43	Remove
Software compatibility	84	2.54	1.43	Remove
Brand	84	3.79	1.02	Retain
After-sales / warranty	84	2.37	1.34	Remove

Note: five-point Likert scale; a criterion is retained when its mean rating is at least 3.0, the pre-committed midpoint threshold. *Source:* authors' own survey data.

in Stage 2. It is worth recording, ahead of the fuller treatment in Chapter 5, that several criteria emphasised in the expert interviews, notably software compatibility, durability, and after-sales support, were screened out by the wider respondent pool, a divergence between professional and end-user perspectives that the two-step design of Stage 1 makes visible rather than concealing.

4.3 Criteria Weights and Consistency Check

The retained criteria were compared pairwise by the representative expert as described in Section 3.3, producing a 7×7 reciprocal matrix from 21 elicited judgements. Before the weights were interpreted, the matrix was submitted to the consistency check of Equation (3.3): the principal eigenvalue is $\lambda_{\text{max}} = 7.3823$, giving a consistency index of $CI = 0.0637$ and, with the random index $RI = 1.32$ for $n = 7$, a consistency ratio of $CR = 0.0483$. As this value lies below the acceptance threshold of 0.10 (Saaty, 2008), the judgement set is internally coherent and the derived weights are usable without re-elicitation. Table 4.3 presents the resulting priority vector.

The weight structure is strongly concentrated at the top. Price carries the largest weight (0.3491) and the processor the second largest (0.2769); together these two criteria account for approximately 62.6% of the total priority, so the model characterises the office-laptop decision primarily as a trade-off between budget and computing capability. A middle tier follows, comprising storage (0.1592) and memory (0.1201), while brand (0.0430), size or weight (0.0258), and battery life (0.0258) form a low-priority tier that collectively holds less than 10% of the weight. The position of battery life is notable: although it recorded the highest mean in the

Table 4.3: Criteria weights obtained by AHP.

Criterion	Weight (eigenvector)
Price	0.3491
CPU	0.2769
RAM	0.1201
SSD	0.1592
Brand	0.0430
Size / weight	0.0258
Battery life	0.0258

Note: weights are the principal eigenvector of the pairwise comparison matrix; $\lambda_{\max} = 7.3823$, $CI = 0.0637$, $CR = 0.0483 < 0.10$. *Source:* authors' own computation.

screening survey, it receives the smallest AHP weight, and the same reversal separates this result from Weng Siew Lam et al. (2023), who report battery life as the most heavily weighted criterion in their mobile-phone model. This pattern is consistent with the recalibration that emerged from the expert interviews, in which battery life is a requirement that most office laptops already satisfy rather than a differentiator; its interpretation is developed in Chapter 5. As a robustness check, the weights were recomputed by the geometric-mean method described in Section 3.3; the resulting vector preserves the same ordering of all seven criteria, indicating that the priority structure does not depend on the derivation method. These weights answer RQ2 and enter Stage 3 as the weighting vector for TOPSIS.

4.4 Decision Scenario and Alternatives

Applying the outlet pool and deduplication procedure of Section 3.4.2 to the seven review outlets of Table 3.3 produced twelve distinct candidate laptops after configuration variants of the same model were collapsed into a single entry. Price and configuration data for all twelve candidates were collected from Amazon.com on 2026-07-12, following the sourcing rationale of Section 3.4.2. To keep the ranking table and the calculation tables of Chapter E compact, each candidate is assigned a short identifier, Can1 through Can12, in the order of collection; Table 4.4 reports the assignment, and the subsequent tables refer to the candidates by these identifiers.

Table 4.5 reports the attribute values in the form in which they were collected, that is, before the CPU and Brand values are converted into the PassMark benchmark score and the Gartner market-share figure described in Section 3.4.3; that converted matrix is the input actually used by the TOPSIS procedure of Section 4.5.

The twelve candidates span an order-of-magnitude price range, from $395 for the entry-level Acer Aspire 5 to $3,499 for the top-specification ThinkPad X1 Carbon Gen 14 Aura Edition, and are drawn from six brands, with Lenovo contributing six of the twelve listings. Every candidate except the Acer Aspire 5 carries 32 GB of RAM or the 24 GB unified-memory configuration used by Apple's M-series chips, so the RAM criterion mainly distinguishes the Aspire 5 and

Table 4.4: Candidate identifiers assigned to the twelve laptop models.

Candidate	Model
Can1	Lenovo ThinkPad X1 Carbon Gen 13 Aura Edition
Can2	Lenovo ThinkPad X1 Carbon Gen 14 Aura Edition
Can3	Apple MacBook Air M4
Can4	Apple MacBook Air M5
Can5	Asus ExpertBook P5
Can6	HP EliteBook Ultra G1i
Can7	Lenovo ThinkPad X9 15 Aura Edition (2025)
Can8	Dell 14 Premium (XPS 14 2026)
Can9	Lenovo ThinkPad T14 Gen 6 (AMD)
Can10	Lenovo ThinkPad T14 Gen 7
Can11	Acer Aspire 5
Can12	Lenovo ThinkPad X1 Carbon Gen 12

Source: authors' own compilation from the review outlets of Table 3.3.

Table 4.5: Candidate laptops and their raw attribute values.

Model	Price	CPU	RAM	SSD	Brand	Weight	Battery
Lenovo ThinkPad X1 Carbon Gen 13 Aura Edition	2289	Intel Core Ultra 7 258V	32	2048	Lenovo	1.09	57
Lenovo ThinkPad X1 Carbon Gen 14 Aura Edition	3499	Intel Core Ultra 7 365 vPro	32	2048	Lenovo	1.00	58
Apple MacBook Air M4	1494	Apple M4 (10-core)	24	512	Apple	1.24	53.8
Apple MacBook Air M5	1749	Apple M5 (10-core)	24	1024	Apple	1.23	53.8
Asus ExpertBook P5	1675	Intel Core Ultra 7 258V	32	1024	ASUS	1.29	63
HP EliteBook Ultra G1i	1539	Intel Core Ultra 7 258V	32	1024	HP	1.36	56
Lenovo ThinkPad X9 15 Aura Edition (2025)	1799	Intel Core Ultra 7 258V	32	1024	Lenovo	1.40	80
Dell 14 Premium (XPS 14 2026)	1890	Intel Core Ultra 7 255H	32	1024	Dell	1.72	69.5
Lenovo ThinkPad T14 Gen 6 (AMD)	1100	AMD Ryzen AI 5 340	16	512	Lenovo	1.41	57
Lenovo ThinkPad T14 Gen 7	2357	Intel Core Ultra 7 355 (Series 3)	32	512	Lenovo	2.22	60
Acer Aspire 5	395	Intel Core i3-1115G4	8	128	Acer	1.63	48
Lenovo ThinkPad X1 Carbon Gen 12	1685	Intel Core Ultra 165U vPro	32	1024	Lenovo	1.08	57

Note: price in USD, RAM and SSD in GB, weight in kg, battery capacity in Wh; attribute values as listed at the point of sale. *Source:* Amazon.com product listings, collected 12 July 2026.

the 16 GB ThinkPad T14 Gen 6 from the rest of the field rather than separating the remaining ten candidates from one another. The benchmark and market-share conversion applied to the CPU and Brand columns, together with the full step-by-step TOPSIS calculation, is reported in Chapter E.

4.5 TOPSIS Ranking Results

Applying the seven-step TOPSIS procedure of Section 3.4.4 to the converted decision matrix of Section E.1, using the AHP weights of Table 4.3, produces the closeness coefficients and ranking reported in Table 4.6. The full normalisation, weighting, and ideal-solution calculations underlying this table are reported step by step in Section E.2.

Table 4.6: TOPSIS ranking of the candidate laptops.

Rank	Candidate	C_i
1	Can9 — Lenovo ThinkPad T14 Gen 6 (AMD)	0.6152
2	Can8 — Dell 14 Premium (XPS 14 2026)	0.6046
3	Can4 — Apple MacBook Air M5	0.6039
4	Can6 — HP EliteBook Ultra G1i	0.5901
5	Can3 — Apple MacBook Air M4	0.5868
6	Can5 — Asus ExpertBook P5	0.5621
7	Can11 — Acer Aspire 5	0.5608
8	Can12 — Lenovo ThinkPad X1 Carbon Gen 12	0.5444
9	Can7 — Lenovo ThinkPad X9 15 Aura Edition (2025)	0.5416
10	Can1 — Lenovo ThinkPad X1 Carbon Gen 13 Aura Edition	0.5170
11	Can10 — Lenovo ThinkPad T14 Gen 7	0.4140
12	Can2 — Lenovo ThinkPad X1 Carbon Gen 14 Aura Edition	0.3803

Note: C_i is the closeness coefficient; candidate identifiers as assigned in Table 4.4. *Source:* authors' own computation.

The Lenovo ThinkPad T14 Gen 6 (AMD) obtains the highest closeness coefficient ($C_i = 0.6152$), ahead of the Dell 14 Premium ($C_i = 0.6046$) and the Apple MacBook Air M5 ($C_i = 0.6039$). The three leading alternatives are separated by no more than 0.011 in closeness coefficient, a narrow margin that leaves the top of the ranking sensitive to small changes in the underlying scores. The top-ranked model reaches this position without leading on any single criterion: its price of $1,100 is the second lowest in the set, its processor score (19,546) is in the middle of the field, and its 16 GB of RAM is the smallest configuration among the eleven non-Acer candidates, but it combines a below-median price with Lenovo's market-share advantage on the Brand criterion, and no single weakness is severe enough to offset that combination. The Dell 14 Premium and the MacBook Air M5 reach second and third place from a different combination: both carry the highest or near-highest processor scores in the field (30,684 and 26,831 respectively), which compensates for a price above the sample median.

The lowest-ranked alternative is the Lenovo ThinkPad X1 Carbon Gen 14 Aura Edition ($C_i = 0.3803$), the most expensive candidate at $3,499, despite carrying the second-highest processor score in the set; because Price and CPU together account for 62.6% of the total criteria weight (Section 4.3), a high score on one of these two criteria does not compensate for an extreme value on the other. The Acer Aspire 5, the cheapest candidate at $395, ranks seventh of twelve ($C_i = 0.5608$) rather than first, even though Price carries the single largest weight in the model: its processor, memory, storage, and brand-share values are the lowest or near-lowest on every one of those criteria, and the combined deficit outweighs its price advantage. Taken together, the two extremes illustrate that no criterion, however heavily weighted, guarantees a favourable rank on its own; TOPSIS rewards a candidate's overall balance across the weighted profile rather than dominance on a single attribute. The practical implication of this result, together with its bearing on the recommended alternative, is developed further in Chapter 5.

Chapter 5

Discussion

5.1 Interpretation of the Criteria Weights

The weight structure reported in Section 4.3 concentrates 62.6% of the total priority in price (0.3491) and processor capability (0.2769), and this concentration is readable directly from the decision context the criteria were calibrated against. In the expert interviews, the purchase budget for an office machine is not a preference to be optimised but a policy set in advance by management or the IT department, so price operates as the framing variable within which every subsequent hardware trade-off is made. The dominant position of price is consistent with the wider consumer-behaviour evidence: Liao and Chuang (2022) find price the single most influential attribute in their conjoint analysis of laptop choice, and Šostar and Ristanović (2023) identify personal budget as the strongest driver among the cultural, social, personal, and psychological factors they weigh. The weight of the processor, in turn, reflects the interviews' consistent redirection of scarce budget towards computing capability: the experts' argument for removing the dedicated graphics card (Section 4.1.2) was precisely that the money it consumes buys more useful performance when spent on a newer-generation CPU, more memory, or faster storage.

The more interesting feature of the weight vector, however, is its bottom tier, because this is where the local calibration departs most visibly from the international literature. Battery life records the highest mean in the screening survey (4.02, Section 4.2) yet receives the joint-smallest AHP weight (0.0258, Section 4.3), and Weng Siew Lam et al. (2023) report battery life as the most heavily weighted criterion in their integrated AHP–TOPSIS model. The apparent contradiction dissolves once the two instruments are read as measuring different things. The Likert screening asks whether a criterion matters at all, and battery life clearly does: a machine that cannot hold a charge through a meeting is unusable. The pairwise comparison instead asks how much of a decisive difference the criterion makes between candidate machines, and here the expert judgement recorded in the interviews applies: mains power is readily available in Vietnamese offices, a runtime of four to seven hours already covers meetings and short mobile work, and most current office laptops meet that bar. Battery life therefore behaves as a threshold requirement, broadly satisfied across the candidate set, rather than as a differentiator, and a rational pairwise comparison assigns differentiating weight elsewhere. The same logic extends to physical size and weight, which the survey ranks joint-second in importance (3.96) but the AHP places in the same low tier (0.0258): the interviews emphasised weight because motorbike commuting makes a heavy machine a daily burden, yet the office segment has converged on 14-inch machines weighing roughly 1.3 to 1.6 kilograms, so the attribute varies too little across realistic candidates to move a pairwise judgement. What the survey measures as salience, the AHP re-expresses as marginal decision value, and criteria whose requirements the market already meets lose weight in the second instrument without losing importance in the first.

A related divergence between the expert and end-user perspectives concerns the criteria that the screening survey removed. Software compatibility was added to the candidate set on the strength of the interviews, in which office workers described a Windows lock-in effect created by accounting packages, internal ERP systems, and the government's customs declaration software (Section 4.1.2), yet the wider respondent pool screened it out, together with durability and after-sales support (Section 4.2). The likely mechanism is that end users experience compatibility as a background condition rather than a choice variable: almost every machine they encounter in a Vietnamese office already runs Windows, so the criterion only becomes visible at the moment a non-Windows device enters consideration, which is exactly the procurement-risk scenario the experts are paid to anticipate. The two-step design of Stage 1 makes this divergence observable instead of averaging it away, and the practical reading is that the screened-out criteria function as eligibility conditions applied before the weighted comparison, not as attributes the weighted comparison should ignore.

One divergence worth flagging concerns the granularity of the Display criterion, which the screening survey of Section 4.2 removed for falling just below the retention threshold (mean $= 2.80$, $SD = 1.32$). Read against the wider literature, this outcome may reflect how the criterion was operationalised rather than a genuine indifference to visual quality among Vietnamese office workers. Sönmez Çakır and Pekkaya (2020) decompose monitor properties into four distinct sub-criteria, resolution, size, touch-screen support, and other usage advantages, and find that these carry markedly different weights within the group, with touch-screen support outweighing resolution. Weng Siew Lam et al. (2023) go further and place screen size and pixel density in two separate higher-level categories, physical properties and technical specification respectively, with screen size ranking second within its category while pixel density ranks last within its own; the two attributes therefore answer different buyer concerns, physical footprint versus visual output quality, rather than a single underlying preference. Maghsoudi et al. (2026) report the starkest contrast: in their sentiment-weighted ranking of sixteen notebook features, display quality places third overall, immediately behind price and design, while size places eleventh, a gap that would be invisible had the two been folded into one composite score. Set beside these findings, the single "Display" item used in the present screening instrument bundled resolution, sharpness, anti-glare coating, and brightness into one rating, plausibly diluting a genuinely important sub-attribute, visual quality, with a less decisive one, raw screen size, and depressing the mean below the retention threshold. This is offered as an interpretive caveat rather than a claim that Vietnamese office workers disregard display quality outright, and it points to criterion granularity, not only criterion choice, as a methodological lever meriting attention in future replications of this instrument.

5.2 Interpretation of the Ranking

The identity of the top-ranked alternative is itself informative about what the calibrated model rewards. The Lenovo ThinkPad T14 Gen 6 (AMD) leads the ranking (Section 4.5) without holding the best value on any single criterion: its price is the second lowest in the set, its processor sits in the middle of the field, and its 16 GB of memory is the smallest configuration among the eleven non-Acer candidates. Its closeness coefficient of 0.6152 is instead the product of a profile with no severe weakness, anchored by a below-median price and supported by the largest brand market share in the set. This is precisely the machine the expert interviews described before any market data had been collected: a current-generation mid-tier processor was judged sufficient for administrative workloads, and the budget saved by not chasing peak specifications was to be preserved rather than redirected into features the office segment does not use. That the data-driven ranking independently converges on this profile is a meaningful internal consistency check between Stage 1's qualitative calibration and Stage 3's quantitative outcome.

The two extremes of the ranking reinforce the same reading from opposite directions. The Acer Aspire 5, the cheapest candidate at $395, finishes seventh rather than first despite price carrying the single largest weight, because its deficits on processor, memory, storage, and brand share accumulate faster than its price advantage compensates. At the other end, the ThinkPad X1 Carbon Gen 14 Aura Edition finishes last at $3,499 despite the second-highest processor score in the set, an outcome that echoes the over-specification argument of Rau and Fang (2018): paying a premium for capability beyond the requirements of the use case destroys rather than creates value once price enters the comparison with its full weight. Between these extremes, the top three alternatives are separated by less than 0.011 in closeness coefficient, so the ranking is better read as producing a short list of near-equivalent recommendations, the T14 Gen 6, the Dell 14 Premium, and the MacBook Air M5, than as declaring a single decisive winner; the Dell and the Apple machine reach their positions through the complementary route of field-leading processor scores that offset above-median prices. Within that short list, the practical tie-breakers are exactly the eligibility conditions identified in Section 5.1: the MacBook's macOS platform, for instance, would be filtered out by the Windows lock-in constraint in offices where that constraint binds, even though the weighted comparison itself ranks it third.

5.3 Practical and Managerial Implications

For an individual office worker buying with personal funds, the model's main practical export is not the specific ranking, which is tied to a catalogue snapshot, but the structure of the weights: roughly 63% of the decision is a price–processor trade-off, storage and memory form a middle tier worth attending to, and battery life, weight, and brand deserve verification against a threshold rather than optimisation. Concretely, the buyer should fix the budget first, secure a current-generation mid-tier processor and 16 GB of memory within it, confirm that battery life and weight fall in the acceptable band for their commuting pattern, and resist paying premiums for capability at the top of the range, which the ranking shows to be counterproductive under office-work weights.

For corporate procurement, the value of the model lies in making an implicit decision auditable. The interviews indicated that purchase budgets are set by policy and that IT managers already think in terms of total cost of ownership; the AHP–TOPSIS pipeline turns those practices into an explicit weight vector and a reproducible ranking that can be attached to a purchase justification, revisited when the catalogue changes, and recalculated by replacing a single input matrix. Because the weights and the alternatives are decoupled, a procurement office can hold the weight vector stable across purchasing rounds while refreshing the decision matrix, which converts a one-off analysis into a reusable procedure. The screened-out criteria retain a role here as eligibility gates: software compatibility and after-sales coverage are best enforced as pass-or-fail conditions before the weighted comparison, mirroring how the experts described actual procurement practice.

The results also carry signals for the supply side. For retailers advising office customers, the ranking suggests that the persuasive assortment is not the cheapest machine or the most powerful one but the balanced mid-tier configuration, and that advice framed as "sufficient processor within budget" matches how this segment actually weighs the purchase. For manufacturers, the weight structure implies that marginal engineering investment aimed at this segment earns more return in price positioning and processor-generation currency than in pushing battery life or thinness beyond the threshold the segment already considers satisfied; a battery lasting beyond the four-to-seven-hour working band, in particular, adds cost against a criterion holding under 3% of the decision weight.

5.4 Comparison with Prior Studies

Set against the prior studies reviewed in Section 2.5, the present results agree on the centre of the priority structure and diverge at its edges, and the divergences are traceable to context rather than method. The dominance of price is the most stable point of agreement: it matches the conjoint evidence of Liao and Chuang (2022), the budget-driven consumer behaviour reported by Šostar and Ristanović (2023), and the price-sensitive student segment of Elnatan and Tannady (n.d.), in which price, processor, and memory jointly lead the weights, a composition strikingly close to the top of the present vector.

The clearest divergence concerns brand. Sönmez Çakır and Pekkaya (2020) find brand image the highest-weighted criterion among their six, whereas the present model assigns brand only 0.0430. The difference is explicable through the signalling mechanism that motivates brand weight in the first place: brand compensates for information asymmetry when buyers cannot evaluate technical quality directly, and that asymmetry is largest for general consumers choosing alone. In the office-procurement context studied here, IT managers and standardised specifications absorb much of that uncertainty, so the signalling value of brand shrinks even while brand retains a residual role, visible in the ranking, where Lenovo's market share contributes to the winning profile. The second divergence, the position of battery life relative to Weng Siew Lam et al. (2023), was examined in Section 5.1 and reflects both the device category, since a phone cannot assume mains power, and the office setting that can.

At the level of ranked outcomes, comparison is necessarily looser because candidate sets differ, but two observations hold. Gaur (2023), ranking laptop brands during the pandemic with a comparable AHP–TOPSIS apparatus, identify Dell as the leading alternative; the present ranking, built on model-level rather than brand-level alternatives, places a Dell machine second, suggesting that the earlier result survives translation to a finer-grained alternative space. Maghsoudi et al. (2026), weighting notebook features by mined sentiment, place the graphics card last among sixteen features and price near the top, which independently corroborates both the experts' removal of the GPU from the candidate set and the weight the present model gives to price; their finding that display quality ranks third, against its removal by the present screening survey, is the one substantive disagreement, and Section 5.1 argues it is most plausibly an artefact of criterion granularity rather than a genuine difference in what office users value. Taken together, the comparisons support the position claimed in Section 2.6: the international literature travels well at the core of the priority structure, price and processing capability, and requires local recalibration precisely at the criteria, battery, weight, brand, and software ecosystem, whose value depends on infrastructure and working conditions.

Chapter 6

Conclusion

6.1 Summary of Findings

This study set out to answer three research questions about how an office worker in Vietnam should select a laptop, and each stage of the three-stage design answers one of them. In answer to RQ1, the combination of literature synthesis, expert interviews, and a screening survey of 84 respondents produced a set of seven evaluation criteria: price, processor capability, memory, solid-state storage, brand, physical size and weight, and battery life. The path to this set is itself part of the finding, since the interviews removed the dedicated graphics card and added software compatibility before the survey screened the candidate list against a pre-committed threshold, leaving a criteria set calibrated to local working conditions rather than imported from the international literature.

In answer to RQ2, the pairwise comparison of the seven criteria produced a consistent judgement matrix ($CR = 0.0483$, below the 0.10 acceptance threshold) whose priority vector is strongly concentrated at the top: price carries a weight of 0.3491 and the processor 0.2769, together accounting for approximately 62.6% of the total priority, followed by storage (0.1592) and memory (0.1201), with brand (0.0430), size and weight (0.0258), and battery life (0.0258) forming a low-priority tier. The most instructive feature of this vector is the position of battery life, which recorded the highest mean in the screening survey yet the smallest pairwise weight, a reversal that Chapter 5 attributes to its status as a threshold requirement that most office laptops already satisfy.

In answer to RQ3, applying TOPSIS with these weights to twelve candidate laptops drawn from seven professional review outlets and priced from Amazon.com ranked the Lenovo ThinkPad T14 Gen 6 (AMD) first with a closeness coefficient of 0.6152, ahead of the Dell 14 Premium (0.6046) and the Apple MacBook Air M5 (0.6039). The recommended machine leads on no single criterion but combines a below-median price, a mid-field processor, and the largest brand market share into the most balanced profile in the set, while the cheapest and the most expensive candidates finish seventh and last respectively, confirming that under the derived weights the office-laptop decision rewards balance rather than dominance on any one attribute.

6.2 Contributions

The contribution of the study is the twofold one stated in Section 1.3, now delivered rather than promised. First, the study constructs a criteria set for office laptops that is explicitly calibrated to Vietnamese working conditions, and documents where that calibration departs from priorities reported in the international literature: battery life recedes from the top-weighted position it holds in prior device-selection studies once mains power can be assumed, device weight is salient for motorbike commuting yet varies too little across the modern office segment to differentiate candidates, and software compatibility emerges as an eligibility condition created by a Windows lock-in effect that international criteria frameworks rarely make explicit. Second, the study operationalises a complete AHP–TOPSIS pipeline on real market data, from elicited pairwise judgements through a consistency-checked weight vector to a ranking of twelve actual laptop models, with every computational step reported in the appendices in reproducible form. Beyond these two, the two-instrument design of Stage 1 carries a methodological point worth stating separately: by screening criteria with end users after eliciting them from experts, the study makes the divergence between professional and end-user perspectives observable, and Chapter 5 shows that this divergence is informative rather than noise.

6.3 Limitations

Four limitations bound the claims made above. First, the pairwise comparison matrix was elicited from a single representative expert, as described in Section 3.3; although the matrix passes the consistency check and the geometric-mean recomputation preserves the same criteria ordering, a weight vector aggregated over several independent experts would carry more evidential force, and the screening survey's divergence between expert and end-user priorities suggests the choice of judge is not neutral. Second, the screening sample of 84 respondents was recruited by convenience rather than probability sampling, so the retained criteria set reflects the office-worker population only to the extent that the sample does. Third, the market data are a snapshot: prices and configurations were collected from a single retail platform on a recorded date (Section 4.4), and laptop prices in particular move quickly enough that the specific ranking, though not the method, has a short shelf life. Fourth, two criteria are measured through proxies, the processor through its PassMark benchmark score and brand through global market share, and while Section 3.4.3 argues these are the most defensible quantitative stand-ins available, neither captures its criterion completely; market share in particular measures prevalence rather than the service quality that motivated the brand criterion in the interviews. In addition, the stability of the ranking under perturbations of the weight vector was not examined, which matters because the top three closeness coefficients lie within 0.011 of one another (Section 4.5); the ranking should accordingly be read as a short list rather than a strict ordering.

6.4 Future Work

The limitations point directly to the extensions worth pursuing. The most immediate is a group-decision version of Stage 2, eliciting pairwise matrices from several experts and aggregating them, which would both strengthen the weight vector and permit an analysis of inter-expert disagreement that the present single-judge design cannot support. A second extension is a formal sensitivity analysis of the ranking with respect to the weights, which the narrow margins at the top of the present ranking make particularly relevant. On the measurement side, fuzzy extensions of AHP and TOPSIS, of the kind reviewed in Sections 2.3 and 2.4, would let the model represent the imprecision of judgements such as brand trust directly instead of forcing them through crisp proxies. On the data side, the pipeline should be re-run as the Vietnamese retail catalogue catches up with the international models considered here, both to replace the Amazon.com price base with domestic prices that include local warranty terms and to test whether the ranking survives the change of market; repeating the collection at intervals would also show how quickly rankings decay as catalogues turn over. Finally, the interpretive caveat raised in Section 5.1 about criterion granularity deserves a direct test: a replication that decomposes the removed Display criterion into visual quality and screen size, following the decomposition used by Sönmez Çakır and Pekkaya (2020), would establish whether office users' apparent indifference to displays is genuine or an artefact of the composite item, and the same design could extend the model to user segments beyond office work, where the calibration developed here is not expected to hold.

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Appendix A

Theoretical Basis for the Candidate Criteria

This appendix sets out the academic and theoretical grounding for the fourteen candidate criteria that literature synthesis (Section 3.2.1) contributed to the candidate set, together with one additional criterion, software compatibility, that was introduced only after the expert interviews of Section 3.2.2. The discussion below is confined to prior research and general consumer-electronics theory; it deliberately excludes the interview-specific reasoning already reported for the Vietnamese context in Section 4.1.2, so that the two sources of justification, literature and fieldwork, remain visibly separate rather than blended into a single narrative.

A.1 Economic Factors and the Budget Constraint

Price occupies a central position in the consumer decision network (Sönmez Çakır & Pekkaya, 2020). Rau and Fang (2018) add an economic framing in which price is treated as compensation for brand value, and device value is understood to depreciate as a function of time. The mechanism behind this depreciation is the pace of technological turnover: a device purchased at a high price can become functionally dated within a few years even without any physical fault, simply because the surrounding software and peripheral ecosystem has moved on. This dynamic explains why, in terms of relative priority, price behaves as a comparatively strong constraint for the general consumer segment (Weng Siew Lam et al., 2023). Price-sensitive buyers are willing to trade away hardware headroom they do not need, for instance accepting a mid-tier processor generation instead of the newest flagship chip, in exchange for a budget that still covers the tasks they actually perform (Maghsoudi et al., 2026). Expert or professional buyers, by contrast, tend to treat price as secondary, accepting a meaningfully higher cost in return for durability and dependable after-sales support. At the opposite end of the market, Jiménez-Parra et al. (2014) find that among buyers of remanufactured electronics a low price is the one incentive strong enough to make a buyer accept the residual risk of using previously owned technology.

A.2 Core Hardware Capability

The processing capacity of the CPU together with the working memory supplied by RAM forms the microarchitectural basis for a computer's execution capability. Because the CPU functions as the bottleneck that governs a device's overall responsiveness, Kang et al. (2022) place it among the core quality attributes, one whose weight in the purchase decision outweighs that of accompanying after-sales services. Examined from a behavioural angle, Liao and Chuang (2022) document a shift in how consumers reason about processors: buyers increasingly favour multi-core architectures chosen not purely to maximise clock speed but to optimise performance per watt of energy consumed, a preference aligned with broader sustainability expectations. Alongside the processor, a large RAM capacity has stopped being an optional upgrade and has instead become a near-mandatory baseline, a shift driven by the requirement to keep several cloud-based platforms and resource-intensive applications running at once. This same shift toward modern hardware also explains why solid-state storage has displaced the traditional hard disk drive: because an SSD contains no spinning platter or other moving mechanical part, it removes the risk of data loss from physical shock while travelling and reduces power draw at the same time.

The graphics processing unit (GPU) exposes a clearer split between customer segments. Users with specialised needs, such as gaming or design work, require strong graphics performance and willingly accept the bulkier, noisier cooling system that a discrete GPU demands. The sentiment analysis of social-media posts conducted by Maghsoudi et al. (2026), however, shows that heat output and fan noise are among the strongest sources of complaint precisely within the office-worker segment. This rejection follows directly from a physical and an economic trade-off. Physically, a discrete GPU requires a larger cooling assembly and draws more power, which increases the device's thickness and weight and shortens battery life, all of which work against the mobility that office use depends on. Economically, because office tasks such as word processing and spreadsheet work cannot make use of a discrete GPU's capacity, fitting one becomes a case of over-specification: it raises the budget constraint discussed in Section A.1 without returning a proportional benefit (Rau & Fang, 2018). These two mechanisms together explain why a component that represents, on paper, superior technical capability is nonetheless declined by the general buyer segment in favour of integrated graphics.

A.3 Portability and Durability

The ability to operate away from a fixed power source makes battery life one of the attributes buyers weigh most heavily when choosing a laptop. Weng Siew Lam et al. (2023) report that, among technical specifications, battery life carries the single highest weight in their model. Buyers oriented toward a more sustainable lifestyle place a comparable emphasis on efficient batteries as a way of lengthening the replacement cycle and limiting electronic waste (Liao & Chuang, 2022). Manufacturers nonetheless face a physical trade-off in delivering this: battery capacity scales with the overall weight of the device.

On design and weight, the evidence runs against the assumption that younger consumers chase a slim form factor regardless of cost. The data reported by Weng Siew Lam et al. (2023) show practicality dominating aesthetics: their student respondents place durability, battery life, and processing power ahead of other attributes, pushing a slim or fashion-oriented design to the bottom of the priority ranking. Zakeri et al. (2023) reinforce this reading by treating durability as the property that determines how far a device's usable lifecycle can be extended. This body of evidence explains why buyers accept a somewhat heavier machine, or pay a premium for an impact-resistant aluminium-alloy shell, in preference to a thin plastic shell that is light but comparatively fragile (Liao & Chuang, 2022).

A.4 Interface and Interaction Experience

Across long working sessions, the display and the keyboard are the two attributes that determine physical comfort and, by extension, how long a session can be sustained without fatigue. Maghsoudi et al. (2026) record that experienced users hold a demanding standard for typing feel, so that a keyboard lacking adequate key travel or tactile feedback draws negative feedback quickly and consistently.

For the display, panel technology matters alongside size. Liao and Chuang (2022) trace the decline of LCD panels to the arrival of LED backlighting, which can cut power consumption by as much as 60 percent, a gain that aligns with the broader move toward more sustainable consumption. Anti-glare coatings and blue-light filtering are likewise becoming a de facto standard, valued for reducing digital eye strain over extended use. Physical connectivity ports follow the opposite trajectory: rather than continuing to be upgraded, they are increasingly folded into a secondary, supplementary feature group (Sönmez Çakır & Pekkaya, 2020), a pattern that reflects the broader shift toward wireless connectivity. The growth of cloud storage together with a maturing ecosystem of Bluetooth peripherals has allowed bulkier physical ports, such as optical drives or wired network sockets, to be removed from many devices altogether, freeing up space for a more minimalist chassis.

A.5 Trust and Emotional Value

Brand is a factor that shapes purchase intention directly, and Sönmez Çakır and Pekkaya (2020) treat it as a filter through which a buyer absorbs the physical and technical properties of a product without evaluating each one individually. Read through a behavioural-economics lens, this filtering role is a response to information asymmetry: because the general buyer lacks the technical background to assess a chip, a cooling assembly, or the durability of a motherboard directly, brand functions as a signalling mechanism that stands in for that missing expertise. Apple is the clearest illustration of the mechanism: office buyers rarely investigate transistor counts or the microarchitecture of an M-series chip, and instead purchase on the strength of a logo that is trusted to guarantee smooth performance, display quality, and battery life. A less established brand offering comparable specifications on paper typically struggles to justify the same price, precisely because it lacks this trust filter to convert raw technical numbers into a sense of purchase safety. Buyers who are strongly attached to a given brand tend, as a consequence, to be comparatively insensitive to price movements elsewhere in the market (Rau & Fang, 2018). This trust function is not always carried by the original manufacturer's logo: in the market for used or remanufactured devices, it is instead the reputation of the distributor or refurbisher that carries this role, countering the buyer's default assumption that a used device is more likely to fail (Jiménez-Parra et al., 2014).

After-sales service is treated by buyers as a form of insurance against risk. Šostar and Ristanović (2023) find that a flexible warranty policy reduces the fear of an expensive component failure and, in doing so, builds a longer-lasting emotional attachment between the buyer and the product. Although after-sales support is often classified as a secondary attribute relative to hardware, Maghsoudi et al. (2026) show that a service failure can trigger a rapid wave of negative sentiment on social media, so that a single after-sales incident is capable of outweighing the goodwill built by genuine hardware improvements.

A.6 Software Compatibility (Added After the Expert Interviews)

Unlike the fourteen criteria discussed above, software compatibility was not part of the literature-derived candidate set; it entered the candidate set only after the expert interviews of Section 3.2.2, for reasons specific to the Vietnamese office context that are reported in Section 4.1.2. Independently of that fieldwork, the wider consumer-electronics literature also treats platform or operating system as a measurable purchase attribute in its own right, rather than a background assumption. Guan and Yuen (2022) build a laptop-recommendation model in which operating system, coded across Linux, OS X, Windows 7, and Windows 8, is included as a standalone attribute, and derive distinct, internally consistent preference values for each option, treating platform choice as directly comparable to attributes such as CPU or RAM rather than as an incidental detail. Weng Siew Lam et al. (2023), summarising a smartphone-selection study by Chakraborty et al., similarly report operating system as the second most heavily weighted criterion overall, behind brand and ahead of user experience, functionality, affordability, and design. Taken together, these two sources provide an independent theoretical basis for treating software or platform compatibility as a legitimate candidate criterion for personal computing devices, a basis that stands apart from, and precedes, the Windows-based accounting and enterprise-software lock-in argument documented from the Vietnamese expert interviews in Section 4.1.2.

A.7 Summary

Table A.1 lists each candidate criterion alongside a short description and the sources on which the discussion above draws.

Table A.1: Candidate criteria and their supporting literature.

Criterion	Description	Sources
Price	Purchase cost of the device	Sönmez Çakır & Pekkaya (2020); Rau & Fang (2018); Jiménez-Parra et al. (2014); Lam et al. (2023); Maghsoudi et al. (2026)
CPU	Processor capability	Kang et al. (2022); Liao & Chuang (2022); Lam et al. (2023); Maghsoudi et al. (2026); Rau & Fang (2018)
RAM	Memory capacity	Sönmez Çakır & Pekkaya (2020); Lam et al. (2023); Maghsoudi et al. (2026); Rau & Fang (2018)
SSD	Storage capacity and type	Sönmez Çakır & Pekkaya (2020); Lam et al. (2023); Maghsoudi et al. (2026); Rau & Fang (2018); Liao & Chuang (2022)
GPU	Graphics processing capability	Maghsoudi et al. (2026); Rau & Fang (2018)
Display	Screen size and panel technology	Sönmez Çakır & Pekkaya (2020); Liao & Chuang (2022); Maghsoudi et al. (2026); Lam et al. (2023)
Keyboard	Typing feel and build quality	Maghsoudi et al. (2026); Liao & Chuang (2022)
Connectivity ports	Range of physical ports	Maghsoudi et al. (2026); Sönmez Çakır & Pekkaya (2020)
Design and shell material	Aesthetics and chassis material	Lam et al. (2023); Kang et al. (2022); Maghsoudi et al. (2026)
Size and weight	Portability	Lam et al. (2023); Zakeri et al. (2023); Maghsoudi et al. (2026)
Battery life	Runtime on battery	Lam et al. (2023); Liao & Chuang (2022); Maghsoudi et al. (2026); Kang et al. (2022); Rau & Fang (2018)
Durability	Shock resistance and build life	Zakeri et al. (2023); Lam et al. (2023); Liao & Chuang (2022)
Brand	Manufacturer reputation	Sönmez Çakır & Pekkaya (2020); Rau & Fang (2018); Jiménez-Parra et al. (2014); Lam et al. (2023)
After-sales	Warranty and customer support	Šostar & Ristanović (2023); Maghsoudi et al. (2026); Kang et al. (2022)
Software compatibility	Operating-system and platform fit *(added after expert interviews)*	Guan & Yuen (2022); Lam et al. (2023)

Source: authors' own synthesis from the studies listed in the Sources column.

Appendix B

Expert Interview Protocol and Summary

B.1 Interview Protocol

The semi-structured interview protocol used with all five experts was constructed following the Interview Protocol Refinement (IPR) framework (Castillo-Montoya, 2016), as summarised in Section 3.2.2. The protocol comprised thirteen core questions organised into the five parts listed in Table B.1, preceded by an informed-consent script explaining the academic purpose of the study, the anonymisation of personal information, and the participant's right to decline any question or withdraw at any time. Each session was expected to last twenty to thirty minutes and was recorded through written notes only, without audio recording.

Part 2 read a short stimulus card to the expert before each question, presenting an academic finding as material for critique rather than as a leading prompt: a card on hardware and performance reported that CPU and RAM were the two criteria rated most important by Maghsoudi et al. (2026), with GPU ranking tenth; a card on mobility and durability reported that Lam et al. (2023) ranked battery life as the most important technical-specification criterion for mobile devices; and a card on experience and trust reported that Sönmez Çakır and Pekkaya (2020) identified brand image as the most important criterion overall, alongside the finding of Maghsoudi et al. (2026) that fan noise and charger bulk drew frequent complaints yet had little influence on the final purchase decision. Most substantive questions carried a follow-up probe asking the respondent to recall the specific case behind an answer or to explain the reasoning process that produced it, a technique adapted from cognitive interviewing (Willis, 2005).

B.2 Expert Profiles

Five experts were selected purposively so that each occupies a different position in the laptop procurement process, as summarised in Table B.2.

Table B.1: Structure of the five-part interview protocol.

Part	Content	Purpose
0	Greeting and informed consent	Establish ethical basis and obtain verbal consent
1 — Warm-up (Q1–Q3)	Recent, concrete laptop-selection episode	Ground the respondent's perspective in a specific case rather than an abstract opinion
2 — Stimulus-based critique (Q4–Q7)	A literature finding for each criterion group is read aloud immediately before the matching question	Anchor agreement or disagreement to a stated academic claim
3 — Vietnamese context (Q8–Q9)	Open questions on locally specific factors	Surface criteria the literature synthesis could not have anticipated
4 — Trade-off scenario (Q10–Q11)	Hypothetical comparisons between conflicting criteria	Generate qualitative seed data for the AHP pairwise comparisons of Section 3.3
5 — Closing (Q12–Q13)	Member-checking and open addition	Confirm the interviewer's understanding and capture unprompted remarks

Source: authors' own protocol design, following Castillo-Montoya (2016) and Willis (2005).

Table B.2: Profiles of the five interviewed experts.

Code	Role	Experience	Position in procurement process	Date
CG-01	Deputy head, international relations office	7 years	End-user with direct purchasing experience	2026-07-05
CG-02	IT administrator	3 years	Specifies and maintains office device fleets	2026-07-09
CG-03	Import–export documentation officer	3.5 years	End-user representing the target persona	2026-07-10
CG-04	Head of product development	10 years	Manager with organisational purchasing authority	2026-07-08
CG-05	Retail sales and technical-support consultant	3 years	Advises laptop buyers daily at point of sale	2026-07-10

Source: authors' own interview data.

B.3 Thematic Summary of Findings

Interview notes were analysed thematically, cross-referencing each recurring position against the two or more experts who independently raised it, consistent with the majority corroboration rule described in Section 3.2.2.

B.3.1 Hardware Priorities and the GPU Question

Four of the five experts converged independently on removing the dedicated graphics card from the criteria set. CG-02 characterised a discrete GPU as over-specification for office work that adds cost, weight, heat, and battery drain without a matching benefit, a position consistent with the tenth-place GPU ranking reported by Maghsoudi et al. (2026) and read to the experts as Card A. CG-03, CG-04, and CG-05 reached the same conclusion from their respective vantage points as a documentation officer, a product-development manager, and a retail consultant, each reporting that office customers who ask about GPU capability are rare and that the budget it would consume is better redirected to the CPU, RAM, or SSD. CG-05 qualified the point by noting that a discrete GPU remains justified only when the organisation has an occasional design or video-editing need, a caveat also raised by CG-02. CG-01 was the sole dissenting voice: in the direct question on GPU priority, CG-01 described integrated graphics as adequate for most office machines but characterised a discrete GPU as "nice to have, no harm in not having it," and in the Part 4 trade-off scenario explicitly chose the GPU-equipped option, reasoning that no buyer can be certain in advance that a device will never be used for design or gaming work, so a more versatile machine is the safer choice. This single dissent is the reason the criteria revision rule in Section 3.2.2 requires majority rather than unanimous corroboration before a candidate criterion is changed. On RAM, all five experts converged on 16 GB as the new practical minimum, citing browser multitasking and simultaneous use of communication platforms such as Microsoft Teams and Zalo (CG-02, CG-03) as the drivers of memory pressure that an 8 GB configuration can no longer absorb.

B.3.2 Battery, Weight, and Design Trade-offs

Four of the five experts (CG-02 through CG-05) explicitly attributed a lower priority on maximum battery capacity to the ready availability of mains power in Vietnamese offices, qualifying the emphasis placed on battery life by Weng Siew Lam et al. (2023), read to the experts as Card B; CG-01 reached the same ranking of priorities without citing mains-power access as the reason. CG-01, CG-03, and CG-05 stated that a runtime of roughly half a working day, sufficient to cover a meeting or a short period away from a desk, is adequate, while CG-04 set a more concrete range of four to seven hours. In its place, physical weight emerged as the more decisive mobility criterion: CG-01, CG-03, CG-04, and CG-05 all linked this preference to motorbike commuting and daily carrying, converging on a practical range of roughly 1.3 to 1.6 kg for a 14-

inch machine. On design, CG-01, CG-03, and CG-05 associated a slim, neutral-coloured form factor with a professional impression in client-facing work, but CG-02 and CG-04 cautioned that this aesthetic preference is bounded by durability requirements, specifically hinge rigidity, thermal headroom, and resistance to chassis flex, so that appearance is treated as a secondary criterion once a minimum durability bar is met.

B.3.3 Brand, After-Sales, and Divergent Views on Trust

Interview responses on brand and after-sales service were the least uniform of the five themes. CG-01 argued that the growing accessibility of AI-assisted comparison tools is reducing buyers' reliance on brand reputation as a proxy for quality, a view echoed by CG-03's observation that end-users increasingly research specifications independently. CG-02, CG-04, and CG-05, however, each reported from a business-procurement or retail vantage point that brand and after-sales reliability remain decisive when device downtime carries an organisational cost: CG-02 emphasised that a business buyer facing two comparably priced machines will choose the more established brand unless an internal IT team can independently verify and service an unfamiliar one, while CG-04 and CG-05 both noted that buyers who cannot evaluate technical specifications in depth default to two or three familiar brands as a risk-management heuristic. CG-05 added that proximity of an authorised service centre to the workplace is frequently a deciding factor for organisational buyers, a point later adopted as the operational rationale for measuring Brand through a service-centre-count proxy in Section 3.4.3.

B.3.4 Locally Specific Factors

All five experts identified factors specific to the Vietnamese working environment that the international literature reviewed in Section 2.5 does not address directly. The hot and humid climate was raised by CG-02 and CG-04 as a driver of thermal-management and build-quality requirements beyond what temperate-climate studies typically consider. Motorbike commuting was raised independently by CG-01, CG-03, CG-04, and CG-05 as the mechanism linking weight and durability to daily use rather than occasional travel. CG-01 and CG-03 both described a software lock-in effect arising from Windows-only accounting, customs-declaration, and internal enterprise-resource-planning software, which excludes non-Windows devices from consideration regardless of hardware merit; this observation directly motivated the addition of software compatibility as a replacement criterion for GPU, as reported in Section 4.1.2. CG-02 and CG-05 further noted that the residual prevalence of HDMI projectors and USB Type-A peripherals in Vietnamese offices keeps traditional ports relevant even as ultraportable designs move toward USB-C only.

B.3.5 Trade-off Scenario Responses

When asked in Part 4 to name the three criteria they would prioritise if all others were kept flexible, core hardware performance (CPU, RAM, SSD, or a combination) appeared in every one of the five answers, making it the only criterion named by all experts. Brand or after-sales support was named by four of the five (CG-02 through CG-05), and durability was named separately by three (CG-02, CG-03, CG-04). Price, although established earlier in the interview as the first screening criterion applied by all five experts, recurred explicitly in only two of the five top-three answers (CG-01, CG-05), suggesting that once a price bracket has already narrowed the option set, experts treat it as a pre-filter rather than a continuing point of comparison. This pattern of convergence on hardware performance as the dominant criterion, together with the comparatively low priority assigned to battery life and to size or weight once the office context was made explicit, foreshadows the weight concentration later confirmed quantitatively by the AHP results in Section 4.3.

Appendix C

Likert Survey Questionnaire

The screening questionnaire described in Section 3.2.3 was administered in Vietnamese through Google Forms; the English rendering below preserves its structure and wording. Respondents rated the importance of each of the fourteen candidate criteria when choosing a laptop for office work on a five-point Likert scale: 1 (not at all important), 2 (not important), 3 (neutral), 4 (important), and 5 (very important). The instrument contained no demographic items.

Table C.1: Items of the criteria-screening questionnaire.

Item	Criterion rated	Scale
1	Price	1–5
2	Processing power (CPU)	1–5
3	Memory capacity (RAM)	1–5
4	Storage space (SSD)	1–5
5	Display	1–5
6	Keyboard experience	1–5
7	Design and shell material	1–5
8	Size and weight (portability)	1–5
9	Durability and shock resistance	1–5
10	Battery life	1–5
11	Connectivity ports	1–5
12	Software compatibility	1–5
13	Brand	1–5
14	After-sales service and warranty	1–5

Note: English rendering of the instrument administered in Vietnamese. *Source:* authors' own questionnaire design.

Appendix D

AHP Pairwise Comparison Matrix and Step-by-Step Calculation

Table D.1 reproduces the full 7×7 pairwise comparison matrix elicited from the representative expert as described in Section 3.3. Entries above the diagonal are the 21 elicited judgements on the Saaty scale; entries below the diagonal follow from reciprocity.

Table D.1: Pairwise comparison matrix of the seven retained criteria.

	Price	**CPU**	**RAM**	**SSD**	**Brand**	**Size/weight**	**Battery**
Price	1	1	5	4	7	8	8
CPU	1	1	3	2	7	9	8
RAM	1/5	1/3	1	1/2	5	6	6
SSD	1/4	1/2	2	1	5	7	6
Brand	1/7	1/7	1/5	1/5	1	2	3
Size/weight	1/8	1/9	1/6	1/7	1/2	1	1
Battery	1/8	1/8	1/6	1/6	1/3	1	1

Note: entries above the diagonal are the 21 elicited judgements on the Saaty scale; entries below the diagonal follow from reciprocity. *Source:* authors' own expert elicitation.

D.1 Step 1 — Column Sums

Each column of the matrix A in Table D.1 is summed to obtain the normalising constant used in the following step. Table D.2 reports the seven column totals.

Table D.2: Column sums of the pairwise comparison matrix.

	Price	**CPU**	**RAM**	**SSD**	**Brand**	**Size/weight**	**Battery**
Column sum	2.8429	3.2123	11.5333	8.0095	25.8333	34.0000	33.0000

Source: authors' own computation.

D.2 Step 2 — Weight Vector by Power Iteration (Eigenvector Method)

Following the justification given in Section 3.3 for using the principal eigenvector rather than the normalised column-average approximation (Ishizaka & Lusti, 2006; Saaty, 2003), the weight vector **w** is obtained by power iteration: starting from the uniform vector $\mathbf{w}^{(0)} = (1/7, \ldots, 1/7)$, each iteration computes $\mathbf{w}^{(k+1)} = A\mathbf{w}^{(k)} / \left\| A\mathbf{w}^{(k)} \right\|_1$ until the vector stabilises. Table D.3 traces the first six iterations; the vector has already converged to four decimal places by the sixth iteration, which is taken as the final weight vector.

Table D.3: Convergence of the power-iteration weight vector across six iterations.

Iteration	Price	CPU	RAM	SSD	Brand	Size/weight	Battery
$k = 1$	0.2871	0.2618	0.1607	0.1837	0.0565	0.0257	0.0246
$k = 2$	0.3550	0.2734	0.1208	0.1621	0.0405	0.0242	0.0240
$k = 3$	0.3522	0.2774	0.1182	0.1581	0.0424	0.0258	0.0259
$k = 4$	0.3488	0.2771	0.1201	0.1591	0.0431	0.0259	0.0259
$k = 5$	0.3490	0.2769	0.1202	0.1593	0.0430	0.0258	0.0258
$k = 6$	0.3491	0.2769	0.1201	0.1593	0.0430	0.0258	0.0258

Source: authors' own computation.

The converged weight vector is therefore $\mathbf{w} = (0.3491,\ 0.2769,\ 0.1201,\ 0.1592,\ 0.0430,\ 0.0258,\ 0.0258)$, matching the values reported in Table 4.3 of Section 4.3.

D.3 Step 3 — Consistency Check

The converged weight vector is substituted back into $A\mathbf{w}$, and each component is divided by the corresponding weight to obtain seven estimates of the principal eigenvalue, as shown in Table D.4. Because A satisfies exact reciprocity and the power iteration has converged, all seven ratios agree, giving $\lambda_{\text{max}} = 7.3823$ directly rather than as an average over inconsistent estimates.

Table D.4: Consistency check: $A\mathbf{w}$ and the ratio $(A\mathbf{w})_i/w_i$.

	Price	CPU	RAM	SSD	Brand	Size/weight	Battery
$(A\mathbf{w})_i$	2.5773	2.0444	0.8865	1.1756	0.3174	0.1903	0.1908
$(A\mathbf{w})_i/w_i$	7.3823	7.3823	7.3823	7.3823	7.3823	7.3823	7.3823

Source: authors' own computation.

With $\lambda_{\text{max}} = 7.3823$ and $n = 7$, the consistency index from Equation (3.3) is

$$CI = \frac{\lambda_{\text{max}} - n}{n - 1} = \frac{7.3823 - 7}{6} = 0.0637.$$

Using the random index $RI = 1.32$ tabulated in Table 3.2 for $n = 7$, the consistency ratio is

$$CR = \frac{CI}{RI} = \frac{0.0637}{1.32} = 0.0483,$$

which is below the acceptance threshold of 0.10 (Saaty, 2008), so the judgement matrix is accepted without re-elicitation, as reported in Section 4.3.

D.4 Robustness Check — Geometric-Mean Weights

As a cross-check against the derivation method rather than against the judgement data itself, the same matrix A is also solved by the geometric-mean-of-rows method: for each row i, $\tilde{w}_i = \left(\prod_{j=1}^{n} a_{ij}\right)^{1/n}$, and the seven values are then normalised to sum to one. Table D.5 compares the two methods.

Table D.5: Eigenvector weights versus geometric-mean weights.

	Price	**CPU**	**RAM**	**SSD**	**Brand**	**Size/weight**	**Battery**
Eigenvector (adopted)	0.3491	0.2769	0.1201	0.1592	0.0430	0.0258	0.0258
Geometric mean	0.3366	0.2882	0.1185	0.1615	0.0429	0.0264	0.0259

Source: authors' own computation.

The two methods rank all seven criteria identically and differ from one another by at most 0.013 in absolute weight, confirming that the priority structure reported in Section 4.3 does not depend on which of the two derivation methods is used.

Appendix E

Laptop Dataset, Benchmarks, and TOPSIS Step-by-Step Calculation

E.1 Benchmark and Market-Share Conversion

Following Section 3.4.3, the CPU column of Table 4.5 is converted from a processor name to its PassMark CPU Mark score, and the Brand column is converted from a manufacturer name to its worldwide PC shipment market share for Q1 2026 as reported by Gartner (Gartner, Inc., 2026). Table E.1 and Table E.2 report the two conversion keys; Table E.3 then reports the full converted decision matrix that is submitted to TOPSIS.

Table E.1: CPU-to-benchmark conversion key.

Processor	PassMark CPU Mark
Intel Core Ultra 7 258V	18,898
Intel Core Ultra 7 365 (vPro)	20,946
Apple M4 (10-core)	23,622
Apple M5 (10-core)	26,831
Intel Core Ultra 7 255H	30,684
AMD Ryzen AI 5 340	19,546
Intel Core Ultra 7 355 (Series 3)	20,452
Intel Core i3-1115G4	5,811
Intel Core Ultra 7 165U (vPro)	16,676

Note: one entry per distinct processor in the candidate set. *Source:* PassMark Software (2026).

E.2 TOPSIS Step-by-Step Calculation

Given the converted decision matrix of Table E.3 and the AHP weight vector of Table 4.3, the seven steps of Section 3.4.4 are carried out as follows.

Table E.2: Brand-to-market-share conversion key.

Brand	Market share (%)
Lenovo	26.5
HP	19.3
Dell	16.5
Apple	10.6
ASUS	6.7
Acer	6.4

Note: worldwide PC shipment market share, first quarter of 2026. *Source:* Gartner, Inc. (2026).

Table E.3: Converted decision matrix submitted to TOPSIS.

Candidate	Price	CPU	RAM	SSD	Brand	Weight	Battery
Can1	2289	18898	32	2048	26.5	1.09	57
Can2	3499	20946	32	2048	26.5	1.00	58
Can3	1494	23622	24	512	10.6	1.24	53.8
Can4	1749	26831	24	1024	10.6	1.23	53.8
Can5	1675	18898	32	1024	6.7	1.29	63
Can6	1539	18898	32	1024	19.3	1.36	56
Can7	1799	18898	32	1024	26.5	1.40	80
Can8	1890	30684	32	1024	16.5	1.72	69.5
Can9	1100	19546	16	512	26.5	1.41	57
Can10	2357	20452	32	512	26.5	2.22	60
Can11	395	5811	8	128	6.4	1.63	48
Can12	1685	16676	32	1024	26.5	1.08	57

Note: CPU as PassMark CPU Mark score (Table E.1); Brand as market share in % (Table E.2); price in USD; RAM and SSD in GB; weight in kg; battery capacity in Wh; candidate identifiers as assigned in Table 4.4. *Source:* authors' own computation.

Step 2 — Vector normalisation. Following Equation (3.4), each column of Table E.3 is divided by the Euclidean norm of that column across the twelve candidates. Table E.4 reports the seven column norms; Table E.5 reports the resulting normalised matrix r_{ij}.

Table E.4: Euclidean norms of the criterion columns.

	Price	**CPU**	**RAM**	**SSD**	**Brand**	**Weight**	**Battery**
Norm	6672.10	72103.18	98.31	3934.83	71.89	4.94	207.70

Note: each norm is $\sqrt{\sum_i x_{ij}^2}$ taken over the twelve candidates. *Source:* authors' own computation.

Table E.5: Normalised decision matrix r_{ij}.

Candidate	Price	CPU	RAM	SSD	Brand	Weight	Battery
Can1	0.3431	0.2621	0.3255	0.5205	0.3686	0.2206	0.2744
Can2	0.5244	0.2905	0.3255	0.5205	0.3686	0.2024	0.2792
Can3	0.2239	0.3276	0.2441	0.1301	0.1474	0.2510	0.2590
Can4	0.2621	0.3721	0.2441	0.2602	0.1474	0.2490	0.2590
Can5	0.2510	0.2621	0.3255	0.2602	0.0932	0.2611	0.3033
Can6	0.2307	0.2621	0.3255	0.2602	0.2684	0.2753	0.2696
Can7	0.2696	0.2621	0.3255	0.2602	0.3686	0.2834	0.3852
Can8	0.2833	0.4256	0.3255	0.2602	0.2295	0.3481	0.3346
Can9	0.1649	0.2711	0.1628	0.1301	0.3686	0.2854	0.2744
Can10	0.3533	0.2836	0.3255	0.1301	0.3686	0.4493	0.2889
Can11	0.0592	0.0806	0.0814	0.0325	0.0890	0.3299	0.2311
Can12	0.2525	0.2313	0.3255	0.2602	0.3686	0.2186	0.2744

Source: authors' own computation.

Step 3 — Weighted normalised matrix. Following Equation (3.5), each column of Table E.5 is multiplied by the corresponding AHP weight from Table 4.3, giving the weighted matrix v_{ij} in Table E.6.

Step 4 — Ideal solutions. Following Equation (3.6), for each of the seven columns of Table E.6, the positive ideal solution A^+ takes the maximum value for a benefit criterion (CPU, RAM, SSD, Brand, Battery) and the minimum for a cost criterion (Price, Weight); A^- takes the opposite extreme in each case. Table E.7 reports both vectors.

Steps 5–7 — Separation measures, closeness coefficient, and ranking. Following Equation (3.7), for each candidate, S_i^+ and S_i^- are the Euclidean distance from its row of Table E.6 to A^+ and A^- respectively. Table E.8 reports both distances for all twelve candidates. Substituting S_i^+ and S_i^- into Equation (3.8) gives the closeness coefficient C_i for each candidate; the resulting values and final ranking are reported in Table 4.6 of Section 4.5.

Table E.6: Weighted normalised decision matrix v_{ij}.

Candidate	Price	CPU	RAM	SSD	Brand	Weight	Battery
Can1	0.11978	0.07258	0.03910	0.08287	0.01585	0.00569	0.00708
Can2	0.18309	0.08045	0.03910	0.08287	0.01585	0.00522	0.00721
Can3	0.07818	0.09073	0.02932	0.02072	0.00634	0.00648	0.00668
Can4	0.09152	0.10305	0.02932	0.04143	0.00634	0.00642	0.00668
Can5	0.08765	0.07258	0.03910	0.04143	0.00401	0.00674	0.00783
Can6	0.08053	0.07258	0.03910	0.04143	0.01154	0.00710	0.00696
Can7	0.09414	0.07258	0.03910	0.04143	0.01585	0.00731	0.00994
Can8	0.09890	0.11785	0.03910	0.04143	0.00987	0.00898	0.00863
Can9	0.05756	0.07507	0.01955	0.02072	0.01585	0.00736	0.00708
Can10	0.12334	0.07855	0.03910	0.02072	0.01585	0.01159	0.00745
Can11	0.02067	0.02232	0.00977	0.00518	0.00383	0.00851	0.00596
Can12	0.08817	0.06405	0.03910	0.04143	0.01585	0.00564	0.00708

Source: authors' own computation.

Table E.7: Positive (A^+) and negative (A^-) ideal solutions.

	Price	**CPU**	**RAM**	**SSD**	**Brand**	**Weight**	**Battery**
A^+	0.02067	0.11785	0.03910	0.08287	0.01585	0.00522	0.00994
A^-	0.18309	0.02232	0.00977	0.00518	0.00383	0.01159	0.00596

Source: authors' own computation.

Table E.8: Separation measures S_i^+ and S_i^- for each candidate.

Candidate	S_i^+	S_i^-
Can1	0.10900	0.11667
Can2	0.16670	0.10228
Can3	0.09002	0.12784
Can4	0.08458	0.12897
Can5	0.09165	0.11763
Can6	0.08591	0.12369
Can7	0.09575	0.11311
Can8	0.08882	0.13579
Can9	0.08631	0.13799
Can10	0.12647	0.08934
Can11	0.12725	0.16245
Can12	0.09579	0.11449

Note: candidate identifiers as assigned in Table 4.4. *Source:* authors' own computation.